1

Measure Theory

In this chapter, we will recall some definitions and results from measure theory. Our purpose here is to provide an introduction to readers who have not seen these concepts before and to review that material for those who have. Harder proofs, especially those that do not contribute much to one's intuition, are hidden away in the Appendix. Readers with a solid background in measure theory can skip Sections 1.4, 1.5, and 1.7, which were previously part of the Appendix.

1.1 Probability Spaces

Here and throughout the book, terms being defined are set in **boldface**. We begin with the most basic quantity. A **probability space** is a triple $(\Omega, \mathcal{F}, P)$, where Ω is a set of "outcomes," $\mathcal{F}$ is a set of "events," and $P : \mathcal{F} \to [0, 1]$ is a function that assigns probabilities to events. We assume that $\mathcal{F}$ is a σ**-field** (or σ**-algebra**), i.e., a (nonempty) collection of subsets of Ω that satisfy

(i) if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$, and

(ii) if $A_i \in \mathcal{F}$ is a countable sequence of sets, then $\cup_i A_i \in \mathcal{F}$.

Here and in what follows, **countable** means finite or countably infinite. Since $\cap_i A_i = (\cup_i A_i^c)^c$, it follows that a σ-field is closed under countable intersections. We omit the last property from the definition to make it easier to check.

Without P, $(\Omega, \mathcal{F})$ is called a **measurable space**, i.e., it is a space on which we can put a measure. A **measure** is a nonnegative countably additive set function; that is, a function $\mu : \mathcal{F} \to \mathbf{R}$ with

(i) $\mu(A) \geq \mu(\emptyset) = 0$ for all $A \in \mathcal{F}$, and

(ii) if $A_i \in \mathcal{F}$ is a countable sequence of disjoint sets, then

$$\mu(\cup_i A_i) = \sum_i \mu(A_i)$$

If $\mu(\Omega) = 1$, we call μ a **probability measure**. In this book, probability measures are usually denoted by P.

The next result gives some consequences of the definition of a measure that we will need later. In all cases, we assume that the sets we mention are in $\mathcal{F}$.

Theorem 1.1.1 *Let μ be a measure on $(\Omega, \mathcal{F})$*

(i) **monotonicity**. *If $A \subset B$, then $\mu(A) \le \mu(B)$.*

(ii) **subadditivity**. *If $A \subset \cup_{m=1}^\infty A_m$, then $\mu(A) \le \sum_{m=1}^\infty \mu(A_m)$.*

(iii) **continuity from below**. *If $A_i \uparrow A$ (i.e., $A_1 \subset A_2 \subset \dots$ and $\cup_i A_i = A$), then $\mu(A_i) \uparrow \mu(A)$.*

(iv) **continuity from above**. *If $A_i \downarrow A$ (i.e., $A_1 \supset A_2 \supset \dots$ and $\cap_i A_i = A$), with $\mu(A_1) < \infty$, then $\mu(A_i) \downarrow \mu(A)$.*

Proof (i) Let $B - A = B \cap A^c$ be the **difference** of the two sets. Using $+$ to denote disjoint union, $B = A + (B - A)$ so

$$\mu(B) = \mu(A) + \mu(B - A) \ge \mu(A).$$

(ii) Let $A'_n = A_n \cap A$, $B_1 = A'_1$ and for $n > 1$, $B_n = A'_n - \cup_{m=1}^{n-1} A'_m$. Since the B_n are disjoint and have union A, we have using (ii) of the definition of measure, $B_m \subset A_m$, and (i) of this theorem

$$\mu(A) = \sum_{m=1}^\infty \mu(B_m) \le \sum_{m=1}^\infty \mu(A_m)$$

(iii) Let $B_n = A_n - A_{n-1}$. Then the B_n are disjoint and have $\cup_{m=1}^\infty B_m = A$, $\cup_{m=1}^n B_m = A_n$ so

$$\mu(A) = \sum_{m=1}^\infty \mu(B_m) = \lim_{n\to\infty} \sum_{m=1}^n \mu(B_m) = \lim_{n\to\infty} \mu(A_n)$$

(iv) $A_1 - A_n \uparrow A_1 - A$ so (iii) implies $\mu(A_1 - A_n) \uparrow \mu(A_1 - A)$. Since $A_1 \supset A$, we have $\mu(A_1 - A) = \mu(A_1) - \mu(A)$ and it follows that $\mu(A_n) \downarrow \mu(A)$. □

The simplest setting, which should be familiar from undergraduate probability, is:

Example 1.1.2 (Discrete probability spaces) Let Ω = a countable set, i.e., finite or countably infinite. Let $\mathcal{F}$ = the set of all subsets of Ω. Let

$$P(A) = \sum_{\omega \in A} p(\omega), \quad \text{where } p(\omega) \ge 0 \text{ and } \sum_{\omega \in \Omega} p(\omega) = 1$$

A little thought reveals that this is the most general probability measure on this space. In many cases when Ω is a finite set, we have $p(\omega) = 1/|\Omega|$, where $|\Omega|$ = the number of points in Ω.

For a simple concrete example that requires this level of generality, consider the astragali, dice used in ancient Egypt made from the ankle bones of sheep. This die could come to rest on the top side of the bone for four points or on the bottom for three points. The side of the bone was slightly rounded. The die could come to rest on a flat and narrow piece for six points or somewhere on the rest of the side for one point. There is no reason to think that all four outcomes are equally likely, so we need probabilities p_1, p_3, p_4, and p_6 to describe P.

To prepare for our next definition, we need to note that it follows easily from the definition: If $\mathcal{F}_i$, $i \in I$ are σ-fields, then $\cap_{i\in I} \mathcal{F}_i$ is. Here $I \ne \emptyset$ is an arbitrary index set (i.e., possibly uncountable). From this it follows that if we are given a set Ω and a collection $\mathcal{A}$ of subsets of Ω, then there is a smallest σ-field containing $\mathcal{A}$. We will call this the **σ-field generated by** $\mathcal{A}$ and denote it by $\sigma(\mathcal{A})$.

Let $\mathbf{R}^d$ be the set of vectors $(x_1, \ldots x_d)$ of real numbers and $\mathcal{R}^d$ be the **Borel sets,** the smallest σ-field containing the open sets. When $d = 1$, we drop the superscript.

Example 1.1.3 (Measures on the real line) Measures on $(\mathbf{R}, \mathcal{R})$ are defined by giving a **Stieltjes measure function** with the following properties:

(i) F is nondecreasing.

(ii) F is right continuous, i.e., $\lim_{y \downarrow x} F(y) = F(x)$.

Theorem 1.1.4 *Associated with each Stieltjes measure function F there is a unique measure μ on $(\mathbf{R}, \mathcal{R})$ with $\mu((a,b]) = F(b) - F(a)$*

$$\mu((a,b]) = F(b) - F(a) \tag{1.1.1}$$

When $F(x) = x$, the resulting measure is called **Lebesgue measure**.

The proof of Theorem 1.1.4 is a long and winding road, so we will content ourselves with describing the main ideas involved in this section and hiding the remaining details in the Appendix in Section A.1. The choice of "closed on the right" in $(a,b]$ is dictated by the fact that if $b_n \downarrow b$, then we have

$$\cap_n (a, b_n] = (a,b]$$

The next definition will explain the choice of "open on the left."

A collection $\mathcal{S}$ of sets is said to be a **semialgebra** if (i) it is closed under intersection, i.e., $S, T \in \mathcal{S}$ implies $S \cap T \in \mathcal{S}$, and (ii) if $S \in \mathcal{S}$, then S^c is a finite disjoint union of sets in $\mathcal{S}$. An important example of a semialgebra is

Example 1.1.5 $\mathcal{S}_d =$ the empty set plus all sets of the form

$$(a_1, b_1] \times \cdots \times (a_d, b_d] \subset \mathbf{R}^d \quad \text{where } -\infty \le a_i < b_i \le \infty$$

The definition in (1.1.1) gives the values of μ on the semialgebra $\mathcal{S}_1$. To go from semialgebra to σ-algebra, we use an intermediate step. A collection $\mathcal{A}$ of subsets of Ω is called an **algebra** (or **field**) if $A, B \in \mathcal{A}$ implies A^c and $A \cup B$ are in $\mathcal{A}$. Since $A \cap B = (A^c \cup B^c)^c$, it follows that $A \cap B \in \mathcal{A}$. Obviously, a σ-algebra is an algebra. An example in which the converse is false is:

Example 1.1.6 Let $\Omega = \mathbf{Z} =$ the integers. $\mathcal{A} =$ the collection of $A \subset \mathbf{Z}$ so that A or A^c is finite is an algebra.

Lemma 1.1.7 *If $\mathcal{S}$ is a semialgebra, then $\bar{\mathcal{S}} = \{$finite disjoint unions of sets in $\mathcal{S}\}$ is an algebra, called the* **algebra generated by** $\mathcal{S}$.

Proof Suppose $A = +_i S_i$ and $B = +_j T_j$, where $+$ denotes disjoint union and we assume the index sets are finite. Then $A \cap B = +_{i,j} S_i \cap T_j \in \bar{\mathcal{S}}$. As for complements, if $A = +_i S_i$ then $A^c = \cap_i S_i^c$. The definition of $\mathcal{S}$ implies $S_i^c \in \bar{\mathcal{S}}$. We have shown that $\bar{\mathcal{S}}$ is closed under intersection, so it follows by induction that $A^c \in \bar{\mathcal{S}}$. □

Example 1.1.8 Let $\Omega = \mathbf{R}$ and $\mathcal{S} = \mathcal{S}_1$ then $\bar{\mathcal{S}}_1 =$ the empty set plus all sets of the form

$$\cup_{i=1}^k (a_i, b_i] \quad \text{where } -\infty \le a_i < b_i \le \infty$$

Given a set function μ on $\mathcal{S}$ we can extend it to $\bar{\mathcal{S}}$ by

$$\mu\left(+_{i=1}^n A_i\right) = \sum_{i=1}^n \mu(A_i)$$

By a **measure on an algebra** $\mathcal{A}$, we mean a set function μ with

(i) $\mu(A) \ge \mu(\emptyset) = 0$ for all $A \in \mathcal{A}$, and

(ii) if $A_i \in \mathcal{A}$ are disjoint *and their union is in* $\mathcal{A}$, then

$$\mu\left(\cup_{i=1}^\infty A_i\right) = \sum_{i=1}^\infty \mu(A_i)$$

μ is said to be σ**-finite** if there is a sequence of sets $A_n \in \mathcal{A}$ so that $\mu(A_n) < \infty$ and $\cup_n A_n = \Omega$. Letting $A_1' = A_1$ and for $n \ge 2$,

$$A_n' = \cup_{m=1}^n A_m \quad \text{or} \quad A_n' = A_n \cap \left(\cap_{m=1}^{n-1} A_m^c\right) \in \mathcal{A}$$

we can without loss of generality assume that $A_n \uparrow \Omega$ or the A_n are disjoint.

The next result helps us to extend a measure defined on a semialgebra $\mathcal{S}$ to the σ-algebra it generates, $\sigma(\mathcal{S})$

Theorem 1.1.9 *Let $\mathcal{S}$ be a semialgebra and let μ defined on $\mathcal{S}$ have $\mu(\emptyset) = 0$. Suppose (i) if $S \in \mathcal{S}$, is a finite disjoint union of sets $S_i \in \mathcal{S}$, then $\mu(S) = \sum_i \mu(S_i)$, and (ii) if $S_i, S \in \mathcal{S}$ with $S = +_{i\ge 1} S_i$, then $\mu(S) \le \sum_{i\ge 1} \mu(S_i)$. Then μ has a unique extension $\bar{\mu}$ that is a measure on $\bar{\mathcal{S}}$, the algebra generated by $\mathcal{S}$. If $\bar{\mu}$ is sigma-finite, then there is a unique extension ν that is a measure on $\sigma(\mathcal{S})$.*

In (ii) above, and in what follows, $i \ge 1$ indicates a countable union, while a plain subscript i or j indicates a finite union. The proof of Theorems 1.1.9 is rather involved so it is given in Section A.1. To check condition (ii) in the theorem the following is useful.

Lemma 1.1.10 *Suppose only that (i) holds.*
(a) If $A, B_i \in \bar{\mathcal{S}}$ with $A = +_{i=1}^n B_i$, then $\bar{\mu}(A) = \sum_i \bar{\mu}(B_i)$.
(b) If $A, B_i \in \bar{\mathcal{S}}$ with $A \subset \cup_{i=1}^n B_i$, then $\bar{\mu}(A) \le \sum_i \bar{\mu}(B_i)$.

Proof Observe that it follows from the definition that if $A = +_i B_i$ is a finite disjoint union of sets in $\bar{\mathcal{S}}$ and $B_i = +_j S_{i,j}$, then

$$\bar{\mu}(A) = \sum_{i,j} \mu(S_{i,j}) = \sum_i \bar{\mu}(B_i)$$

To prove (b), we begin with the case $n = 1$, $B_1 = B$. $B = A + (B \cap A^c)$ and $B \cap A^c \in \bar{\mathcal{S}}$, so

$$\bar{\mu}(A) \le \bar{\mu}(A) + \bar{\mu}(B \cap A^c) = \bar{\mu}(B)$$

To handle $n > 1$ now, let $F_k = B_1^c \cap \ldots \cap B_{k-1}^c \cap B_k$ and note

$$\cup_i B_i = F_1 + \cdots + F_n$$
$$A = A \cap (\cup_i B_i) = (A \cap F_1) + \cdots + (A \cap F_n)$$

so using (a), (b) with $n = 1$, and (a) again

$$\bar{\mu}(A) = \sum_{k=1}^{n} \bar{\mu}(A \cap F_k) \leq \sum_{k=1}^{n} \bar{\mu}(F_k) = \bar{\mu}\left(\cup_i B_i\right) \qquad \square$$

Proof of Theorem 1.1.4. Let $\mathcal{S}$ be the semialgebra of half-open intervals $(a,b]$ with $-\infty \leq a < b \leq \infty$. To define μ on $\mathcal{S}$, we begin by observing that

$$F(\infty) = \lim_{x \uparrow \infty} F(x) \quad \text{and} \quad F(-\infty) = \lim_{x \downarrow -\infty} F(x) \quad \text{exist}$$

and $\mu((a,b]) = F(b) - F(a)$ makes sense for all $-\infty \leq a < b \leq \infty$ since $F(\infty) > -\infty$ and $F(-\infty) < \infty$.

If $(a,b] = +_{i=1}^{n}(a_i, b_i]$, then after relabeling the intervals we must have $a_1 = a$, $b_n = b$, and $a_i = b_{i-1}$ for $2 \leq i \leq n$, so condition (i) in Theorem 1.1.9 holds. To check (ii), suppose first that $-\infty < a < b < \infty$, and $(a,b] \subset \cup_{i \geq 1}(a_i, b_i]$ where (without loss of generality) $-\infty < a_i < b_i < \infty$. Pick $\delta > 0$ so that $F(a + \delta) < F(a) + \epsilon$ and pick η_i so that

$$F(b_i + \eta_i) < F(b_i) + \epsilon 2^{-i}$$

The open intervals $(a_i, b_i + \eta_i)$ cover $[a + \delta, b]$, so there is a finite subcover (α_j, β_j), $1 \leq j \leq J$. Since $(a + \delta, b] \subset \cup_{j=1}^{J}(\alpha_j, \beta_j]$, (b) in Lemma 1.1.10 implies

$$F(b) - F(a + \delta) \leq \sum_{j=1}^{J} F(\beta_j) - F(\alpha_j) \leq \sum_{i=1}^{\infty} (F(b_i + \eta_i) - F(a_i))$$

So, by the choice of δ and η_i,

$$F(b) - F(a) \leq 2\epsilon + \sum_{i=1}^{\infty} (F(b_i) - F(a_i))$$

and since ϵ is arbitrary, we have proved the result in the case $-\infty < a < b < \infty$. To remove the last restriction, observe that if $(a,b] \subset \cup_i (a_i, b_i]$ and $(A, B] \subset (a,b]$ has $-\infty < A < B < \infty$, then we have

$$F(B) - F(A) \leq \sum_{i=1}^{\infty} (F(b_i) - F(a_i))$$

Since the last result holds for any finite $(A, B] \subset (a,b]$, the desired result follows. $\square$

Measures on $\mathbf{R}^d$

Our next goal is to prove a version of Theorem 1.1.4 for $\mathbf{R}^d$. The first step is to introduce the assumptions on the defining function F. By analogy with the case $d = 1$ it is natural to assume:

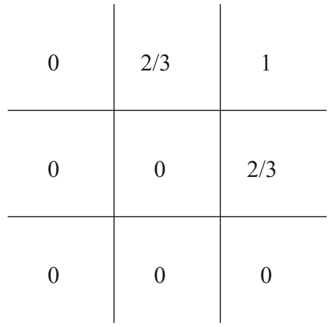

Figure 1.1 Picture of the counterexample.

(i) It is nondecreasing, i.e., if $x \le y$ (meaning $x_i \le y_i$ for all i), then $F(x) \le F(y)$.

(ii) F is right continuous, i.e., $\lim_{y \downarrow x} F(y) = F(x)$ (here $y \downarrow x$ means each $y_i \downarrow x_i$).

(iii) If $x_n \downarrow -\infty$, i.e., each coordinate does, then $F(x_n) \downarrow 0$. If $x_n \uparrow -\infty$, i.e., each coordinate does, then $F(x_n) \uparrow 1$.

However, this time it is not enough. Consider the following F

$$F(x_1, x_2) = \begin{cases} 1 & \text{if } x_1, x_2 \ge 1 \\ 2/3 & \text{if } x_1 \ge 1 \text{ and } 0 \le x_2 < 1 \\ 2/3 & \text{if } x_2 \ge 1 \text{ and } 0 \le x_1 < 1 \\ 0 & \text{otherwise} \end{cases}$$

See Figure 1.1 for a picture. A little thought shows that

$$\begin{aligned} \mu((a_1, b_1] \times (a_2, b_2]) &= \mu((-\infty, b_1] \times (-\infty, b_2]) - \mu((-\infty, a_1] \times (-\infty, b_2]) \\ &\quad - \mu((-\infty, b_1] \times (-\infty, a_2]) + \mu((-\infty, a_1] \times (-\infty, a_2]) \\ &= F(b_1, b_2) - F(a_1, b_2) - F(b_1, a_2) + F(a_1, a_2) \end{aligned}$$

Using this with $a_1 = a_2 = 1 - \epsilon$ and $b_1 = b_2 = 1$ and letting $\epsilon \to 0$ we see that

$$\mu(\{1, 1\}) = 1 - 2/3 - 2/3 + 0 = -1/3$$

Similar reasoning shows that $\mu(\{1, 0\}) = \mu(\{0, 1\}) = 2/3$.

To formulate the third and final condition for F to define a measure, let

$$\begin{aligned} A &= (a_1, b_1] \times \cdots \times (a_d, b_d] \\ V &= \{a_1, b_1\} \times \cdots \times \{a_d, b_d\} \end{aligned}$$

where $-\infty < a_i < b_i < \infty$. To emphasize that ∞'s are not allowed, we will call A a finite rectangle. Then $V =$ the vertices of the rectangle A. If $v \in V$, let

$$\begin{aligned} \text{sgn}\,(v) &= (-1)^{\#\text{ of } a\text{'s in } v} \\ \Delta_A F &= \sum_{v \in V} \text{sgn}\,(v) F(v) \end{aligned}$$

We will let $\mu(A) = \Delta_A F$, so we must assume

(iv) $\Delta_A F \geq 0$ for all rectangles A.

Theorem 1.1.11 *Suppose $F : \mathbf{R}^d \to [0,1]$ satisfies (i)–(iv) given above. Then there is a unique probability measure μ on $(\mathbf{R}^d, \mathcal{R}^d)$ so that $\mu(A) = \Delta_A F$ for all finite rectangles.*

Example 1.1.12 Suppose $F(x) = \prod_{i=1}^d F_i(x)$, where the F_i satisfy (i) and (ii) of Theorem 1.1.4. In this case,

$$\Delta_A F = \prod_{i=1}^d (F_i(b_i) - F_i(a_i))$$

When $F_i(x) = x$ for all i, the resulting measure is Lebesgue measure on $\mathbf{R}^d$.

Proof We let $\mu(A) = \Delta_A F$ for all finite rectangles and then use monotonicity to extend the definition to $\mathcal{S}_d$. To check (i) of Theorem 1.1.9, call $A = +_k B_k$ a **regular subdivision** of A if there are sequences $a_i = \alpha_{i,0} < \alpha_{i,1} \ldots < \alpha_{i,n_i} = b_i$ so that each rectangle B_k has the form

$$(\alpha_{1,j_1-1}, \alpha_{1,j_1}] \times \cdots \times (\alpha_{d,j_d-1}, \alpha_{d,j_d}] \quad \text{where } 1 \leq j_i \leq n_i$$

It is easy to see that for regular subdivisions $\lambda(A) = \sum_k \lambda(B_k)$. (First consider the case in which all the endpoints are finite and then take limits to get the general case.) To extend this result to a general finite subdivision $A = +_j A_j$, subdivide further to get a regular one.

The proof of (ii) is almost identical to that in Theorem 1.1.4. To make things easier to write and to bring out the analogies with Theorem 1.1.4, we let

$$(x,y) = (x_1,y_1) \times \cdots \times (x_d,y_d)$$
$$(x,y] = (x_1,y_1] \times \cdots \times (x_d,y_d]$$
$$[x,y] = [x_1,y_1] \times \cdots \times [x_d,y_d]$$

for $x, y \in \mathbf{R}^d$. Suppose first that $-\infty < a < b < \infty$, where the inequalities mean that each component is finite, and suppose $(a,b] \subset \cup_{i \geq 1}(a^i, b^i]$, where (without loss of generality) $-\infty < a^i < b^i < \infty$. Let $\bar{1} = (1, \ldots, 1)$, pick $\delta > 0$ so that

$$\mu((a - \delta\bar{1}, b]) > \mu((a,b]) - \epsilon$$

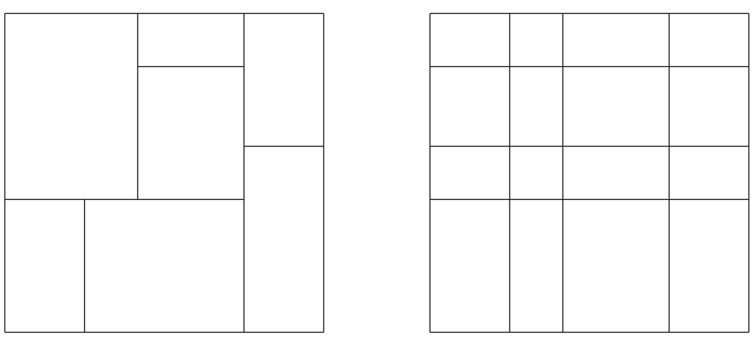

Figure 1.2 Conversion of a subdivision to a regular one.

and pick η_i so that

$$\mu((a, b^i + \eta_i \bar{1}]) < \mu((a^i, b^i]) + \epsilon 2^{-i}$$

The open rectangles $(a^i, b^i + \eta_i \bar{1})$ cover $[a + \delta\bar{1}, b]$, so there is a finite subcover (α^j, β^j), $1 \le j \le J$. Since $(a + \delta\bar{1}, b] \subset \cup_{j=1}^J (\alpha^j, \beta^j]$, (b) in Lemma 1.1.10 implies

$$\mu([a + \delta\bar{1}, b]) \le \sum_{j=1}^{J} \mu((\alpha^j, \beta^j]) \le \sum_{i=1}^{\infty} \mu((a^i, b^i + \eta_i \bar{1}])$$

So, by the choice of δ and η_i,

$$\mu((a, b]) \le 2\epsilon + \sum_{i=1}^{\infty} \mu((a^i, b^i])$$

and since ϵ is arbitrary, we have proved the result in the case $-\infty < a < b < \infty$. The proof can now be completed exactly as before. □

Exercises

1.1.1 Let $\Omega = \mathbf{R}$, $\mathcal{F}$ = all subsets so that A or A^c is countable, $P(A) = 0$ in the first case and $= 1$ in the second. Show that $(\Omega, \mathcal{F}, P)$ is a probability space.

1.1.2 Recall the definition of $\mathcal{S}_d$ from Example 1.1.5. Show that $\sigma(\mathcal{S}_d) = \mathcal{R}^d$, the Borel subsets of $\mathbf{R}^d$.

1.1.3 A σ-field $\mathcal{F}$ is said to be **countably generated** if there is a countable collection $\mathcal{C} \subset \mathcal{F}$ so that $\sigma(\mathcal{C}) = \mathcal{F}$. Show that $\mathcal{R}^d$ is countably generated.

1.1.4 (i) Show that if $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \ldots$ are σ-algebras, then $\cup_i \mathcal{F}_i$ is an algebra. (ii) Give an example to show that $\cup_i \mathcal{F}_i$ need not be a σ-algebra.

1.1.5 A set $A \subset \{1, 2, \ldots\}$ is said to have **asymptotic density** θ if

$$\lim_{n \to \infty} |A \cap \{1, 2, \ldots, n\}|/n = \theta$$

Let $\mathcal{A}$ be the collection of sets for which the asymptotic density exists. Is $\mathcal{A}$ a σ-algebra? an algebra?

1.2 Distributions

Probability spaces become a little more interesting when we define random variables on them. A real-valued function X defined on Ω is said to be a **random variable** if for every Borel set $B \subset \mathbf{R}$ we have $X^{-1}(B) = \{\omega : X(\omega) \in B\} \in \mathcal{F}$. When we need to emphasize the σ-field, we will say that X is $\mathcal{F}$**-measurable** or write $X \in \mathcal{F}$. If Ω is a discrete probability space (see Example 1.1.2), then any function $X : \Omega \to \mathbf{R}$ is a random variable. A second trivial, but useful, type of example of a random variable is the **indicator function** of a set $A \in \mathcal{F}$:

$$1_A(\omega) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \end{cases}$$

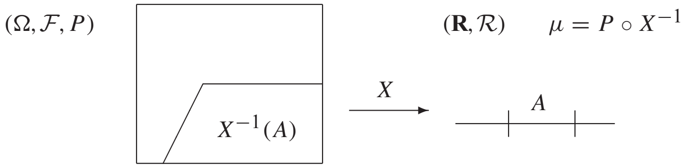

Figure 1.3 Definition of the distribution of X.

The notation is supposed to remind you that this function is 1 on A. Analysts call this object the characteristic function of A. In probability, that term is used for something quite different. (See Section 3.3.)

If X is a random variable, then X induces a probability measure on $\mathbf{R}$ called its **distribution** by setting $\mu(A) = P(X \in A)$ for Borel sets A. Using the notation introduced previously, the right-hand side can be written as $P(X^{-1}(A))$. In words, we pull $A \in \mathcal{R}$ back to $X^{-1}(A) \in \mathcal{F}$ and then take P of that set.

To check that μ is a probability measure we observe that if the A_i are disjoint, then using the definition of μ; the fact that X lands in the union if and only if it lands in one of the A_i; the fact that if the sets $A_i \in \mathcal{R}$ are disjoint, then the events $\{X \in A_i\}$ are disjoint; and the definition of μ again; we have:

$$\mu\left(\cup_i A_i\right) = P\left(X \in \cup_i A_i\right) = P\left(\cup_i \{X \in A_i\}\right) = \sum_i P(X \in A_i) = \sum_i \mu(A_i)$$

The distribution of a random variable X is usually described by giving its **distribution function**, $F(x) = P(X \le x)$.

Theorem 1.2.1 *Any distribution function F has the following properties:*

(i) F is nondecreasing.

(ii) $\lim_{x\to\infty} F(x) = 1$, $\lim_{x\to-\infty} F(x) = 0$.

(iii) F is right continuous, i.e., $\lim_{y\downarrow x} F(y) = F(x)$.

(iv) If $F(x-) = \lim_{y\uparrow x} F(y)$, then $F(x-) = P(X < x)$.

(v) $P(X = x) = F(x) - F(x-)$.

Proof To prove (i), note that if $x \le y$, then $\{X \le x\} \subset \{X \le y\}$, and then use (i) in Theorem 1.1.1 to conclude that $P(X \le x) \le P(X \le y)$.

To prove (ii), we observe that if $x \uparrow \infty$, then $\{X \le x\} \uparrow \Omega$, while if $x \downarrow -\infty$, then $\{X \le x\} \downarrow \emptyset$ and then use (iii) and (iv) of Theorem 1.1.1.

To prove (iii), we observe that if $y \downarrow x$, then $\{X \le y\} \downarrow \{X \le x\}$.

To prove (iv), we observe that if $y \uparrow x$, then $\{X \le y\} \uparrow \{X < x\}$.

For (v), note $P(X = x) = P(X \le x) - P(X < x)$ and use (iii) and (iv). □

The next result shows that we have found more than enough properties to characterize distribution functions.

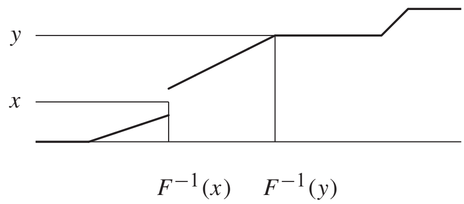

Figure 1.4 Picture of the inverse defined in the proof of Theorem 1.2.2.

Theorem 1.2.2 *If F satisfies (i), (ii), and (iii) in Theorem 1.2.1, then it is the distribution function of some random variable.*

Proof Let $\Omega = (0,1)$, $\mathcal{F}$ = the Borel sets, and P = Lebesgue measure. If $\omega \in (0,1)$, let

$$X(\omega) = \sup\{y : F(y) < \omega\}$$

Once we show that

$$(\star) \qquad \{\omega : X(\omega) \le x\} = \{\omega : \omega \le F(x)\}$$

the desired result follows immediately since $P(\omega : \omega \le F(x)) = F(x)$. (Recall P is Lebesgue measure.) To check $(\star)$, we observe that if $\omega \le F(x)$, then $X(\omega) \le x$, since $x \notin \{y : F(y) < \omega\}$. On the other hand if $\omega > F(x)$, then since F is right continuous, there is an $\epsilon > 0$ so that $F(x+\epsilon) < \omega$ and $X(\omega) \ge x + \epsilon > x$. □

Even though F may not be 1-1 and onto we will call X the inverse of F and denote it by F^{-1}. The scheme in the proof of Theorem 1.2.2 is useful in generating random variables on a computer. Standard algorithms generate random variables U with a uniform distribution, then one applies the inverse of the distribution function defined in Theorem 1.2.2 to get a random variable $F^{-1}(U)$ with distribution function F.

If X and Y induce the same distribution μ on $(\mathbf{R}, \mathcal{R})$, we say X and Y are **equal in distribution**. In view of Theorem 1.1.4, this holds if and only if X and Y have the same distribution function, i.e., $P(X \le x) = P(Y \le x)$ for all x. When X and Y have the same distribution, we like to write

$$X \stackrel{d}{=} Y$$

but this is too tall to use in text, so for typographical reasons we will also use $X =_d Y$.

When the distribution function $F(x) = P(X \le x)$ has the form

$$F(x) = \int_{-\infty}^{x} f(y)\, dy \tag{1.2.1}$$

we say that X has **density function** f. In remembering formulas, it is often useful to think of $f(x)$ as being $P(X = x)$ although

$$P(X = x) = \lim_{\epsilon \to 0} \int_{x-\epsilon}^{x+\epsilon} f(y)\, dy = 0$$

By popular demand, we have ceased our previous practice of writing $P(X = x)$ for the density function. Instead we will use things like the lovely and informative $f_X(x)$.

We can start with f and use (1.2.1) to define a distribution function F. In order to end up with a distribution function, it is necessary and sufficient that $f(x) \geq 0$ and $\int f(x)\,dx = 1$. Three examples that will be important in what follows are:

Example 1.2.3 (Uniform distribution on (0,1)) $f(x) = 1$ for $x \in (0, 1)$ and 0 otherwise. Distribution function:

$$F(x) = \begin{cases} 0 & x \leq 0 \\ x & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

Example 1.2.4 (Exponential distribution with rate λ) $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ and 0 otherwise. Distribution function:

$$F(x) = \begin{cases} 0 & x \leq 0 \\ 1 - e^{-\lambda x} & x \geq 0 \end{cases}$$

Example 1.2.5 (Standard normal distribution)

$$f(x) = (2\pi)^{-1/2} \exp(-x^2/2)$$

In this case, there is no closed form expression for $F(x)$, but we have the following bounds that are useful for large x:

Theorem 1.2.6 *For $x > 0$,*

$$(x^{-1} - x^{-3}) \exp(-x^2/2) \leq \int_x^\infty \exp(-y^2/2)dy \leq x^{-1} \exp(-x^2/2)$$

Proof Changing variables $y = x + z$ and using $\exp(-z^2/2) \leq 1$ gives

$$\int_x^\infty \exp(-y^2/2)\,dy \leq \exp(-x^2/2) \int_0^\infty \exp(-xz)\,dz = x^{-1} \exp(-x^2/2)$$

For the other direction, we observe

$$\int_x^\infty (1 - 3y^{-4}) \exp(-y^2/2)\,dy = (x^{-1} - x^{-3}) \exp(-x^2/2)$$ □

A distribution function on **R** is said to be **absolutely continuous** if it has a density, and **singular** if the corresponding measure is singular w.r.t. Lebesgue measure. See Section A.4 for more on these notions. An example of a singular distribution is:

Example 1.2.7 (Uniform distribution on the Cantor set) The Cantor set C is defined by removing $(1/3, 2/3)$ from [0,1] and then removing the middle third of each interval that remains. We define an associated distribution function by setting $F(x) = 0$ for $x \leq 0$, $F(x) = 1$ for $x \geq 1$, $F(x) = 1/2$ for $x \in [1/3, 2/3]$, $F(x) = 1/4$ for $x \in [1/9, 2/9]$, $F(x) = 3/4$ for $x \in [7/9, 8/9]$, ... Then extend F to all of $[0, 1]$ using monotonicity. There is no f for which (1.2.1) holds because such an f would be equal to 0 on a set of measure 1. From the definition, it is immediate that the corresponding measure has $\mu(C^c) = 0$.

A probability measure P (or its associated distribution function) is said to be **discrete** if there is a countable set S with $P(S^c) = 0$. The simplest example of a discrete distribution is

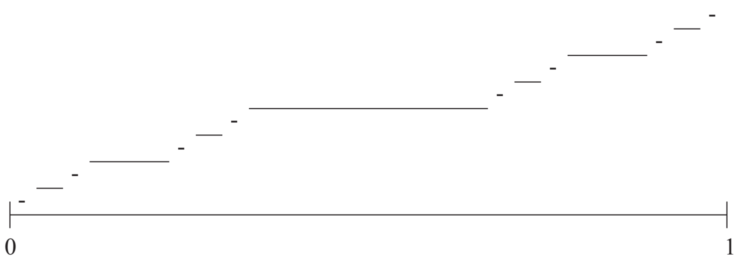

Figure 1.5 Cantor distribution function.

Example 1.2.8 (Point mass at 0) $F(x) = 1$ for $x \geq 0$, $F(x) = 0$ for $x < 0$.

In Section 1.6, we will see the Bernoulli, Poisson, and geometric distributions. The next example shows that the distribution function associated with a discrete probability measure can be quite wild.

Example 1.2.9 (Dense discontinuities) Let $q_1, q_2 \ldots$ be an enumeration of the rationals. Let $\alpha_i > 0$ have $\sum_{i=1}^{\infty} \alpha_1 = 1$ and let

$$F(x) = \sum_{i=1}^{\infty} \alpha_i 1_{[q_i,\infty)}$$

where $1_{[\theta,\infty)}(x) = 1$ if $x \in [\theta, \infty) = 0$ otherwise.

Exercises

1.2.1 Suppose X and Y are random variables on $(\Omega, \mathcal{F}, P)$ and let $A \in \mathcal{F}$. Show that if we let $Z(\omega) = X(\omega)$ for $\omega \in A$ and $Z(\omega) = Y(\omega)$ for $\omega \in A^c$, then Z is a random variable.

1.2.2 Let χ have the standard normal distribution. Use Theorem 1.2.6 to get upper and lower bounds on $P(\chi \geq 4)$.

1.2.3 Show that a distribution function has at most countably many discontinuities.

1.2.4 Show that if $F(x) = P(X \leq x)$ is continuous, then $Y = F(X)$ has a uniform distribution on (0,1), that is, if $y \in [0, 1]$, $P(Y \leq y) = y$.

1.2.5 Suppose X has continuous density f, $P(\alpha \leq X \leq \beta) = 1$ and g is a function that is strictly increasing and differentiable on (α, β). Then $g(X)$ has density $f(g^{-1}(y))/g'(g^{-1}(y))$ for $y \in (g(\alpha), g(\beta))$ and 0 otherwise. When $g(x) = ax + b$ with $a > 0$, $g^{-1}(y) = (y - b)/a$ so the answer is $(1/a)f((y - b)/a)$.

1.2.6 Suppose X has a normal distribution. Use the previous exercise to compute the density of $\exp(X)$. (The answer is called the **lognormal distribution**.)

1.2.7 (i) Suppose X has density function f. Compute the distribution function of X^2 and then differentiate to find its density function. (ii) Work out the answer when X has a standard normal distribution to find the density of the **chi-square distribution**.

1.3 Random Variables

In this section, we will develop some results that will help us later to prove that quantities we define are random variables, i.e., they are measurable. Since most of what we have to say is true for random elements of an arbitrary measurable space $(S, \mathcal{S})$ and the proofs are the same (sometimes easier), we will develop our results in that generality. First we need a definition. A function $X : \Omega \to S$ is said to be a **measurable map** from $(\Omega, \mathcal{F})$ to $(S, \mathcal{S})$ if

$$X^{-1}(B) \equiv \{\omega : X(\omega) \in B\} \in \mathcal{F} \quad \text{for all } B \in \mathcal{S}$$

If $(S, \mathcal{S}) = (\mathbf{R}^d, \mathcal{R}^d)$ and $d > 1$, then X is called a **random vector**. Of course, if $d = 1$, X is called a **random variable**, or r.v. for short.

The next result is useful for proving that maps are measurable.

Theorem 1.3.1 *If $\{\omega : X(\omega) \in A\} \in \mathcal{F}$ for all $A \in \mathcal{A}$ and $\mathcal{A}$* **generates** *$\mathcal{S}$ (i.e., $\mathcal{S}$ is the smallest σ-field that contains $\mathcal{A}$), then X is measurable.*

Proof Writing $\{X \in B\}$ as shorthand for $\{\omega : X(\omega) \in B\}$, we have

$$\{X \in \cup_i B_i\} = \cup_i \{X \in B_i\}$$
$$\{X \in B^c\} = \{X \in B\}^c$$

So the class of sets $\mathcal{B} = \{B : \{X \in B\} \in \mathcal{F}\}$ is a σ-field. Since $\mathcal{B} \supset \mathcal{A}$ and $\mathcal{A}$ generates $\mathcal{S}$, $\mathcal{B} \supset \mathcal{S}$. □

It follows from the two equations displayed in the previous proof that if $\mathcal{S}$ is a σ-field, then $\{\{X \in B\} : B \in \mathcal{S}\}$ is a σ-field. It is the smallest σ-field on Ω that makes X a measurable map. It is called the **σ-field generated by** X and denoted $\sigma(X)$. For future reference we note that

$$\sigma(X) = \{\{X \in B\} : B \in \mathcal{S}\} \tag{1.3.1}$$

Example 1.3.2 If $(S, \mathcal{S}) = (\mathbf{R}, \mathcal{R})$, then possible choices of $\mathcal{A}$ in Theorem 1.3.1 are $\{(-\infty, x] : x \in \mathbf{R}\}$ or $\{(-\infty, x) : x \in \mathbf{Q}\}$ where $\mathbf{Q} =$ the rationals.

Example 1.3.3 If $(S, \mathcal{S}) = (\mathbf{R}^d, \mathcal{R}^d)$, a useful choice of $\mathcal{A}$ is

$$\{(a_1, b_1) \times \cdots \times (a_d, b_d) : -\infty < a_i < b_i < \infty\}$$

or occasionally the larger collection of open sets.

Theorem 1.3.4 *If $X : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ and $f : (S, \mathcal{S}) \to (T, \mathcal{T})$ are measurable maps, then $f(X)$ is a measurable map from $(\Omega, \mathcal{F})$ to $(T, \mathcal{T})$*

Proof Let $B \in \mathcal{T}$. $\{\omega : f(X(\omega)) \in B\} = \{\omega : X(\omega) \in f^{-1}(B)\} \in \mathcal{F}$, since by assumption $f^{-1}(B) \in \mathcal{S}$. □

From Theorem 1.3.4, it follows immediately that if X is a random variable, then so is cX for all $c \in \mathbf{R}$, X^2, $\sin(X)$, etc. The next result shows why we wanted to prove Theorem 1.3.4 for measurable maps.

Theorem 1.3.5 *If $X_1, \ldots X_n$ are random variables and $f : (\mathbf{R}^n, \mathcal{R}^n) \to (\mathbf{R}, \mathcal{R})$ is measurable, then $f(X_1, \ldots, X_n)$ is a random variable.*

Proof In view of Theorem 1.3.4, it suffices to show that $(X_1, \dots, X_n)$ is a random vector. To do this, we observe that if $A_1, \dots, A_n$ are Borel sets, then

$$\{(X_1, \dots, X_n) \in A_1 \times \dots \times A_n\} = \cap_i \{X_i \in A_i\} \in \mathcal{F}$$

Since sets of the form $A_1 \times \dots \times A_n$ generate $\mathcal{R}^n$, the desired result follows from Theorem 1.3.1. □

Theorem 1.3.6 *If $X_1, \dots, X_n$ are random variables, then $X_1 + \dots + X_n$ is a random variable.*

Proof In view of Theorem 1.3.5 it suffices to show that $f(x_1, \dots, x_n) = x_1 + \dots + x_n$ is measurable. To do this, we use Example 1.3.2 and note that $\{x : x_1 + \dots + x_n < a\}$ is an open set and hence is in $\mathcal{R}^n$. □

Theorem 1.3.7 *If $X_1, X_2, \dots$ are random variables then so are*

$$\inf_n X_n \qquad \sup_n X_n \qquad \limsup_n X_n \qquad \liminf_n X_n$$

Proof Since the infimum of a sequence is $< a$ if and only if some term is $< a$ (if all terms are $\geq a$ then the infimum is), we have

$$\{\inf_n X_n < a\} = \cup_n \{X_n < a\} \in \mathcal{F}$$

A similar argument shows $\{\sup_n X_n > a\} = \cup_n \{X_n > a\} \in \mathcal{F}$. For the last two, we observe

$$\liminf_{n\to\infty} X_n = \sup_n \left(\inf_{m\geq n} X_m \right)$$

$$\limsup_{n\to\infty} X_n = \inf_n \left(\sup_{m\geq n} X_m \right)$$

To complete the proof in the first case, note that $Y_n = \inf_{m\geq n} X_m$ is a random variable for each n so $\sup_n Y_n$ is as well. □

From Theorem 1.3.7, we see that

$$\Omega_o \equiv \{\omega : \lim_{n\to\infty} X_n \text{ exists }\} = \{\omega : \limsup_{n\to\infty} X_n - \liminf_{n\to\infty} X_n = 0\}$$

is a measurable set. (Here $\equiv$ indicates that the first equality is a definition.) If $P(\Omega_o) = 1$, we say that X_n **converges almost surely**, or a.s. for short. This type of convergence is called **almost everywhere** in measure theory. To have a limit defined on the whole space, it is convenient to let

$$X_\infty = \limsup_{n\to\infty} X_n$$

but this random variable may take the value $+\infty$ or $-\infty$. To accommodate this and some other headaches, we will generalize the definition of random variable.

A function whose domain is a set $D \in \mathcal{F}$ and whose range is $\mathbf{R}^* \equiv [-\infty, \infty]$ is said to be a **random variable** if for all $B \in \mathcal{R}^*$ we have $X^{-1}(B) = \{\omega : X(\omega) \in B\} \in \mathcal{F}$. Here $\mathcal{R}^*$ = the Borel subsets of $\mathbf{R}^*$ with $\mathbf{R}^*$ given the usual topology, i.e., the one generated by intervals of the form $[-\infty, a)$, (a, b) and $(b, \infty]$ where $a, b \in \mathbf{R}$. The reader should note that the **extended real line** $(\mathbf{R}^*, \mathcal{R}^*)$ is a measurable space, so all the results above generalize immediately.

Exercises

1.3.1 Show that if $\mathcal{A}$ generates $\mathcal{S}$, then $X^{-1}(\mathcal{A}) \equiv \{\{X \in A\} : A \in \mathcal{A}\}$ generates $\sigma(X) = \{\{X \in B\} : B \in \mathcal{S}\}$.

1.3.2 Prove Theorem 1.3.6 when $n = 2$ by checking $\{X_1 + X_2 < x\} \in \mathcal{F}$.

1.3.3 Show that if f is continuous and $X_n \to X$ almost surely then $f(X_n) \to f(X)$ almost surely.

1.3.4 (i) Show that a continuous function from $\mathbf{R}^d \to \mathbf{R}$ is a measurable map from $(\mathbf{R}^d, \mathcal{R}^d)$ to $(\mathbf{R}, \mathcal{R})$. (ii) Show that $\mathcal{R}^d$ is the smallest σ-field that makes all the continuous functions measurable.

1.3.5 A function f is said to be **lower semicontinuous** or l.s.c. if

$$\liminf_{y \to x} f(y) \geq f(x)$$

and **upper semicontinuous** (u.s.c.) if $-f$ is l.s.c. Show that f is l.s.c. if and only if $\{x : f(x) \leq a\}$ is closed for each $a \in \mathbf{R}$ and conclude that semicontinuous functions are measurable.

1.3.6 Let $f : \mathbf{R}^d \to \mathbf{R}$ be an arbitrary function and let $f^\delta(x) = \sup\{f(y) : |y - x| < \delta\}$ and $f_\delta(x) = \inf\{f(y) : |y - x| < \delta\}$ where $|z| = (z_1^2 + \cdots + z_d^2)^{1/2}$. Show that f^δ is l.s.c. and f_δ is u.s.c. Let $f^0 = \lim_{\delta \downarrow 0} f^\delta$, $f_0 = \lim_{\delta \downarrow 0} f_\delta$, and conclude that the set of points at which f is discontinuous $= \{f^0 \neq f_0\}$ is measurable.

1.3.7 A function $\varphi : \Omega \to \mathbf{R}$ is said to be **simple** if

$$\varphi(\omega) = \sum_{m=1}^{n} c_m 1_{A_m}(\omega)$$

where the c_m are real numbers and $A_m \in \mathcal{F}$. Show that the class of $\mathcal{F}$ measurable functions is the smallest class containing the simple functions and closed under pointwise limits.

1.3.8 Use the previous exercise to conclude that Y is measurable with respect to $\sigma(X)$ if and only if $Y = f(X)$ where $f : \mathbf{R} \to \mathbf{R}$ is measurable.

1.3.9 To get a constructive proof of the last result, note that $\{\omega : m2^{-n} \leq Y < (m+1)2^{-n}\} = \{X \in B_{m,n}\}$ for some $B_{m,n} \in \mathcal{R}$ and set $f_n(x) = m2^{-n}$ for $x \in B_{m,n}$ and show that as $n \to \infty$ $f_n(x) \to f(x)$ and $Y = f(X)$.

1.4 Integration

Let μ be a σ-finite measure on $(\Omega, \mathcal{F})$. We will be primarily interested in the special case μ is a probability measure, but we will sometimes need to integrate with respect to infinite measure and it is no harder to develop the results in general.

In this section we will define $\int f \, d\mu$ for a class of measurable functions. This is a four-step procedure:

1. Simple functions
2. Bounded functions
3. Nonnegative functions
4. General functions

This sequence of four steps is also useful in proving integration formulas. See, for example, the proofs of Theorems 1.6.9 and 1.7.2.

Step 1. φ is said to be a **simple function** if $\varphi(\omega) = \sum_{i=1}^n a_i 1_{A_i}$ and A_i are disjoint sets with $\mu(A_i) < \infty$. If φ is a simple function, we let

$$\int \varphi \, d\mu = \sum_{i=1}^n a_i \mu(A_i)$$

The representation of φ is not unique since we have not supposed that the a_i are distinct. However, it is easy to see that the last definition does not contradict itself.

We will prove the next three conclusions four times, but before we can state them for the first time, we need a definition. $\varphi \geq \psi$ μ**-almost everywhere** (or $\varphi \geq \psi$ μ-a.e.) means $\mu(\{\omega : \varphi(\omega) < \psi(\omega)\}) = 0$. When there is no doubt about what measure we are referring to, we drop the μ.

Lemma 1.4.1 *Let φ and ψ be simple functions.*
(i) If $\varphi \geq 0$ a.e. then $\int \varphi \, d\mu \geq 0$.
(ii) For any $a \in \mathbf{R}$, $\int a\varphi \, d\mu = a \int \varphi \, d\mu$.
(iii) $\int \varphi + \psi \, d\mu = \int \varphi \, d\mu + \int \psi \, d\mu$.

Proof (i) and (ii) are immediate consequences of the definition. To prove (iii), suppose

$$\varphi = \sum_{i=1}^m a_i 1_{A_i} \quad \text{and} \quad \psi = \sum_{j=1}^n b_j 1_{B_j}$$

To make the supports of the two functions the same, we let $A_0 = \cup_i B_i - \cup_i A_i$, let $B_0 = \cup_i A_i - \cup_i B_i$, and let $a_0 = b_0 = 0$. Now

$$\varphi + \psi = \sum_{i=0}^m \sum_{j=0}^n (a_i + b_j) 1_{(A_i \cap B_j)}$$

and the $A_i \cap B_j$ are pairwise disjoint, so

$$\begin{aligned}
\int (\varphi + \psi) \, d\mu &= \sum_{i=0}^m \sum_{j=0}^n (a_i + b_j) \mu(A_i \cap B_j) \\
&= \sum_{i=0}^m \sum_{j=0}^n a_i \mu(A_i \cap B_j) + \sum_{j=0}^n \sum_{i=0}^m b_j \mu(A_i \cap B_j) \\
&= \sum_{i=0}^m a_i \, \mu(A_i) + \sum_{j=0}^n b_j \, \mu(B_j) = \int \varphi \, d\mu + \int \psi \, d\mu
\end{aligned}$$

In the next-to-last step, we used $A_i = +_j (A_i \cap B_j)$ and $B_j = +_i (A_i \cap B_j)$, where $+$ denotes a disjoint union. □

We will prove (i)–(iii) three more times as we generalize our integral. As a consequence of (i)–(iii), we get three more useful properties. To keep from repeating their proofs, which do not change, we will prove

Lemma 1.4.2 *If (i) and (iii) hold, then we have:*
(iv) If $\varphi \le \psi$ a.e. then $\int \varphi\, d\mu \le \int \psi\, d\mu$.
(v) If $\varphi = \psi$ a.e. then $\int \varphi\, d\mu = \int \psi\, d\mu$.
If, in addition, (ii) holds when $a = -1$, we have
(vi) $|\int \varphi\, d\mu| \le \int |\varphi|\, d\mu$

Proof By (iii), $\int \psi\, d\mu = \int \varphi\, d\mu + \int (\psi - \varphi)\, d\mu$ and the second integral is ≥ 0 by (i), so (iv) holds. $\varphi = \psi$ a.e. implies $\varphi \le \psi$ a.e. and $\psi \le \varphi$ a.e. so (v) follows from two applications of (iv). To prove (vi) now, notice that $\varphi \le |\varphi|$ so (iv) implies $\int \varphi\, d\mu \le \int |\varphi|\, d\mu$. $-\varphi \le |\varphi|$, so (iv) and (ii) imply $-\int \varphi\, d\mu \le \int |\varphi|\, d\mu$. Since $|y| = \max(y, -y)$, the result follows. □

Step 2. Let E be a set with $\mu(E) < \infty$ and let f be a bounded function that vanishes on E^c. To define the integral of f, we observe that if φ, ψ are simple functions that have $\varphi \le f \le \psi$, then we want to have

$$\int \varphi\, d\mu \le \int f\, d\mu \le \int \psi\, d\mu$$

so we let

$$\int f\, d\mu = \sup_{\varphi \le f} \int \varphi\, d\mu = \inf_{\psi \ge f} \int \psi\, d\mu \tag{1.4.1}$$

Here and for the rest of Step 2, we assume that φ and ψ vanish on E^c. To justify the definition, we have to prove that the sup and inf are equal. It follows from (iv) in Lemma 1.4.2 that

$$\sup_{\varphi \le f} \int \varphi\, d\mu \le \inf_{\psi \ge f} \int \psi\, d\mu$$

To prove the other inequality, suppose $|f| \le M$ and let

$$E_k = \left\{ x \in E : \frac{kM}{n} \ge f(x) > \frac{(k-1)M}{n} \right\} \quad \text{for } -n \le k \le n$$

$$\psi_n(x) = \sum_{k=-n}^{n} \frac{kM}{n} 1_{E_k} \qquad \varphi_n(x) = \sum_{k=-n}^{n} \frac{(k-1)M}{n} 1_{E_k}$$

By definition, $\psi_n(x) - \varphi_n(x) = (M/n)1_E$, so

$$\int \psi_n(x) - \varphi_n(x)\, d\mu = \frac{M}{n}\mu(E)$$

Since $\varphi_n(x) \le f(x) \le \psi_n(x)$, it follows from (iii) in Lemma 1.4.1 that

$$\begin{aligned} \sup_{\varphi \le f} \int \varphi\, d\mu \ge \int \varphi_n\, d\mu &= -\frac{M}{n}\mu(E) + \int \psi_n\, d\mu \\ &\ge -\frac{M}{n}\mu(E) + \inf_{\psi \ge f} \int \psi\, d\mu \end{aligned}$$

The last inequality holds for all n, so the proof is complete. □

Lemma 1.4.3 *Let E be a set with $\mu(E) < \infty$. If f and g are bounded functions that vanish on E^c, then:*

(i) If $f \geq 0$ a.e. then $\int f\, d\mu \geq 0$.
(ii) For any $a \in \mathbf{R}$, $\int af\, d\mu = a \int f\, d\mu$.
(iii) $\int f + g\, d\mu = \int f\, d\mu + \int g\, d\mu$.
(iv) If $g \leq f$ a.e. then $\int g\, d\mu \leq \int f\, d\mu$.
(v) If $g = f$ a.e. then $\int g\, d\mu = \int f\, d\mu$.
(vi) $|\int f\, d\mu| \leq \int |f|\, d\mu$.

Proof Since we can take $\varphi \equiv 0$, (i) is clear from the definition. To prove (ii), we observe that if $a > 0$, then $a\varphi \leq af$ if and only if $\varphi \leq f$, so

$$\int af\, d\mu = \sup_{\varphi \leq f} \int a\varphi\, d\mu = \sup_{\varphi \leq f} a \int \varphi\, d\mu = a \sup_{\varphi \leq f} \int \varphi\, d\mu = a \int f\, d\mu$$

For $a < 0$, we observe that $a\varphi \leq af$ if and only if $\varphi \geq f$, so

$$\int af\, d\mu = \sup_{\varphi \geq f} \int a\varphi\, d\mu = \sup_{\varphi \geq f} a \int \varphi\, d\mu = a \inf_{\varphi \geq f} \int \varphi\, d\mu = a \int f\, d\mu$$

To prove (iii), we observe that if $\psi_1 \geq f$ and $\psi_2 \geq g$, then $\psi_1 + \psi_2 \geq f + g$ so

$$\inf_{\psi \geq f+g} \int \psi\, d\mu \leq \inf_{\psi_1 \geq f, \psi_2 \geq g} \int \psi_1 + \psi_2\, d\mu$$

Using linearity for simple functions, it follows that

$$\begin{aligned} \int f + g\, d\mu &= \inf_{\psi \geq f+g} \int \psi\, d\mu \\ &\leq \inf_{\psi_1 \geq f, \psi_2 \geq g} \int \psi_1\, d\mu + \int \psi_2\, d\mu = \int f\, d\mu + \int g\, d\mu \end{aligned}$$

To prove the other inequality, observe that the last conclusion applied to $-f$ and $-g$ and (ii) imply

$$-\int f + g\, d\mu \leq -\int f\, d\mu - \int g\, d\mu$$

(iv)–(vi) follow from (i)–(iii) by Lemma 1.4.2. □

Notation. We define the integral of f over the set E:

$$\int_E f\, d\mu \equiv \int f \cdot 1_E\, d\mu$$

Step 3. If $f \geq 0$, then we let

$$\int f\, d\mu = \sup\left\{\int h\, d\mu : 0 \leq h \leq f, h \text{ is bounded and } \mu(\{x : h(x) > 0\}) < \infty\right\}$$

The last definition is nice since it is clear that this is well defined. The next result will help us compute the value of the integral.

Lemma 1.4.4 *Let $E_n \uparrow \Omega$ have $\mu(E_n) < \infty$ and let $a \wedge b = \min(a,b)$. Then*

$$\int_{E_n} f \wedge n \, d\mu \uparrow \int f \, d\mu \quad \text{as } n \uparrow \infty$$

Proof It is clear from (iv) in Lemma 1.4.3 that the left-hand side increases as n does. Since $h = (f \wedge n)1_{E_n}$ is a possibility in the sup, each term is smaller than the integral on the right. To prove that the limit is $\int f \, d\mu$, observe that if $0 \le h \le f$, $h \le M$, and $\mu(\{x : h(x) > 0\}) < \infty$, then for $n \ge M$ using $h \le M$, (iv), and (iii),

$$\int_{E_n} f \wedge n \, d\mu \ge \int_{E_n} h \, d\mu = \int h \, d\mu - \int_{E_n^c} h \, d\mu$$

Now $0 \le \int_{E_n^c} h \, d\mu \le M\mu(E_n^c \cap \{x : h(x) > 0\}) \to 0$ as $n \to \infty$, so

$$\liminf_{n \to \infty} \int_{E_n} f \wedge n \, d\mu \ge \int h \, d\mu$$

which proves the desired result since h is an arbitrary member of the class that defines the integral of f. □

Lemma 1.4.5 *Suppose $f, g \ge 0$.*

(i) $\int f \, d\mu \ge 0$
(ii) If $a > 0$, then $\int af \, d\mu = a \int f \, d\mu$.
(iii) $\int f + g \, d\mu = \int f \, d\mu + \int g \, d\mu$
(iv) If $0 \le g \le f$ a.e. then $\int g \, d\mu \le \int f \, d\mu$.
(v) If $0 \le g = f$ a.e. then $\int g \, d\mu = \int f \, d\mu$.

Here we have dropped (vi) because it is trivial for $f \ge 0$.

Proof (i) is trivial from the definition. (ii) is clear, since when $a > 0$, $ah \le af$ if and only if $h \le f$ and we have $\int ah \, d\mu = a \int h \, du$ for h in the defining class. For (iii), we observe that if $f \ge h$ and $g \ge k$, then $f + g \ge h + k$ so taking the sup over h and k in the defining classes for f and g gives

$$\int f + g \, d\mu \ge \int f \, d\mu + \int g \, d\mu$$

To prove the other direction, we observe $(a + b) \wedge n \le (a \wedge n) + (b \wedge n)$ so (iv) from Lemma 1.4.3 and (iii) from Lemma 1.4.4 imply

$$\int_{E_n} (f + g) \wedge n \, d\mu \le \int_{E_n} f \wedge n \, d\mu + \int_{E_n} g \wedge n \, d\mu$$

Letting $n \to \infty$ and using Lemma 1.4.4 gives (iii). As before, (iv) and (v) follow from (i), (iii), and Lemma 1.4.2. □

Step 4. We say f is integrable if $\int |f| \, d\mu < \infty$. Let

$$f^+(x) = f(x) \vee 0 \quad \text{and} \quad f^-(x) = (-f(x)) \vee 0$$

where $a \vee b = \max(a,b)$. Clearly,

$$f(x) = f^+(x) - f^-(x) \quad \text{and} \quad |f(x)| = f^+(x) + f^-(x)$$

We define the integral of f by

$$\int f\,d\mu = \int f^+\,d\mu - \int f^-\,d\mu$$

The right-hand side is well defined since $f^+, f^- \le |f|$ and we have (iv) in Lemma 1.4.5. For the final time, we will prove our six properties. To do this, it is useful to know:

Lemma 1.4.6 *If $f = f_1 - f_2$ where $f_1, f_2 \ge 0$ and $\int f_i\,d\mu < \infty$, then*

$$\int f\,d\mu = \int f_1\,d\mu - \int f_2\,d\mu$$

Proof $f_1 + f^- = f_2 + f^+$ and all four functions are ≥ 0, so by (iii) of Lemma 1.4.5,

$$\int f_1\,d\mu + \int f^-\,d\mu = \int f_1 + f^-\,d\mu = \int f_2 + f^+\,d\mu = \int f_2\,d\mu + \int f^+\,d\mu$$

Rearranging gives the desired conclusion. □

Theorem 1.4.7 *Suppose f and g are integrable.*
(i) If $f \ge 0$ a.e. then $\int f\,d\mu \ge 0$.
(ii) For all $a \in \mathbf{R}$, $\int af\,d\mu = a\int f\,d\mu$.
(iii) $\int f + g\,d\mu = \int f\,d\mu + \int g\,d\mu$
(iv) If $g \le f$ a.e. then $\int g\,d\mu \le \int f\,d\mu$.
(v) If $g = f$ a.e. then $\int g\,d\mu = \int f\,d\mu$.
(vi) $|\int f\,d\mu| \le \int |f|\,d\mu$.

Proof (i) is trivial. (ii) is clear since if $a > 0$, then $(af)^+ = a(f^+)$, and so on. To prove (iii), observe that $f+g = (f^+ + g^+) - (f^- + g^-)$, so using Lemma 1.4.6 and Lemma 1.4.5

$$\begin{aligned}\int f + g\,d\mu &= \int f^+ + g^+\,d\mu - \int f^- + g^-\,d\mu \\ &= \int f^+\,d\mu + \int g^+\,d\mu - \int f^-\,d\mu - \int g^-\,d\mu\end{aligned}$$

As usual, (iv)–(vi) follow from (i)–(iii) and Lemma 1.4.2. □

Notation for special cases:

(a) When $(\Omega, \mathcal{F}, \mu) = (\mathbf{R}^d, \mathcal{R}^d, \lambda)$, we write $\int f(x)\,dx$ for $\int f\,d\lambda$.

(b) When $(\Omega, \mathcal{F}, \mu) = (\mathbf{R}, \mathcal{R}, \lambda)$ and $E = [a, b]$, we write $\int_a^b f(x)\,dx$ for $\int_E f\,d\lambda$.

(c) When $(\Omega, \mathcal{F}, \mu) = (\mathbf{R}, \mathcal{R}, \mu)$ with $\mu((a, b]) = G(b) - G(a)$ for $a < b$, we write $\int f(x)\,dG(x)$ for $\int f\,d\mu$.

(d) When Ω is a countable set, $\mathcal{F}$ = all subsets of Ω, and μ is counting measure, we write $\sum_{i\in\Omega} f(i)$ for $\int f\,d\mu$.

We mention example (d) primarily to indicate that results for sums follow from those for integrals. The notation for the special case in which μ is a probability measure will be taken up in Section 1.6.

Exercises

1.4.1 Show that if $f \geq 0$ and $\int f \, d\mu = 0$, then $f = 0$ a.e.

1.4.2 Let $f \geq 0$ and $E_{n,m} = \{x : m/2^n \leq f(x) < (m+1)/2^n\}$. As $n \uparrow \infty$,

$$\sum_{m=1}^{\infty} \frac{m}{2^n} \mu(E_{n,m}) \uparrow \int f \, d\mu$$

1.4.3 Let g be an integrable function on $\mathbf{R}$ and $\epsilon > 0$. (i) Use the definition of the integral to conclude there is a simple function $\varphi = \sum_k b_k 1_{A_k}$ with $\int |g - \varphi| \, dx < \epsilon$. (ii) Use Exercise A.2.1 to approximate the A_k by finite unions of intervals to get a **step function**

$$q = \sum_{j=1}^{k} c_j 1_{(a_{j-1}, a_j)}$$

with $a_0 < a_1 < \ldots < a_k$, so that $\int |\varphi - q| < \epsilon$. (iii) Round the corners of q to get a continuous function r so that $\int |q - r| \, dx < \epsilon$.

(iv) To make a continuous function replace each $c_j 1_{(a_{j-1}, a_j)}$ by a function that is 0 $(a_{j-1}, a_j)^c$, c_j on $[a_{j-1} + \delta - j, a_j - \delta_j]$, and linear otherwise. If the δ_j are small enough and we let $r(x) = \sum_{j=1}^{k} r_j(x)$, then

$$\int |q(x) - r(x)| \, d\mu = \sum_{j=1}^{k} \delta_j c_j < \epsilon$$

1.4.4 Prove the **Riemann-Lebesgue lemma.** If g is integrable, then

$$\lim_{n \to \infty} \int g(x) \cos nx \, dx = 0$$

Hint: If g is a step function, this is easy. Now use the previous exercise.

1.5 Properties of the Integral

In this section, we will develop properties of the integral defined in the last section. Our first result generalizes (vi) from Theorem 1.4.7.

Theorem 1.5.1 (Jensen's inequality) *Suppose φ is convex, that is,*

$$\lambda\varphi(x) + (1 - \lambda)\varphi(y) \geq \varphi(\lambda x + (1 - \lambda)y)$$

for all $\lambda \in (0, 1)$ and $x, y \in \mathbf{R}$. If μ is a probability measure, and f and $\varphi(f)$ are integrable, then

$$\varphi\left(\int f \, d\mu\right) \leq \int \varphi(f) \, d\mu$$

Proof Let $c = \int f \, d\mu$ and let $\ell(x) = ax + b$ be a linear function that has $\ell(c) = \varphi(c)$ and $\varphi(x) \geq \ell(x)$. To see that such a function exists, recall that convexity implies

$$\lim_{h \downarrow 0} \frac{\varphi(c) - \varphi(c - h)}{h} \leq \lim_{h \downarrow 0} \frac{\varphi(c + h) - \varphi(c)}{h}$$

(The limits exist since the sequences are monotone.) If we let a be any number between the two limits and let $\ell(x) = a(x-c) + \varphi(c)$, then ℓ has the desired properties. With the existence of ℓ established, the rest is easy. (iv) in Theorem 1.4.7 implies

$$\int \varphi(f)\, d\mu \geq \int (af + b)\, d\mu = a \int f\, d\mu + b = \ell\left(\int f\, d\mu\right) = \varphi\left(\int f\, d\mu\right)$$

since $c = \int f\, d\mu$ and $\ell(c) = \varphi(c)$. □

Let $\|f\|_p = (\int |f|^p\, d\mu)^{1/p}$ for $1 \leq p < \infty$, and notice $\|cf\|_p = |c| \cdot \|f\|_p$ for any real number c.

Theorem 1.5.2 (Hölder's inequality) *If $p, q \in (1,\infty)$ with $1/p + 1/q = 1$, then*

$$\int |fg|\, d\mu \leq \|f\|_p \|g\|_q$$

Proof If $\|f\|_p$ or $\|g\|_q = 0$, then $|fg| = 0$ a.e., so it suffices to prove the result when $\|f\|_p$ and $\|g\|_q > 0$ or by dividing both sides by $\|f\|_p\|g\|_q$, when $\|f\|_p = \|g\|_q = 1$. Fix $y \geq 0$ and let

$$\varphi(x) = x^p/p + y^q/q - xy \quad \text{for } x \geq 0$$
$$\varphi'(x) = x^{p-1} - y \quad \text{and} \quad \varphi''(x) = (p-1)x^{p-2}$$

so φ has a minimum at $x_o = y^{1/(p-1)}$. $q = p/(p-1)$ and $x_o^p = y^{p/(p-1)} = y^q$ so

$$\varphi(x_o) = y^q(1/p + 1/q) - y^{1/(p-1)}y = 0$$

Since x_o is the minimum, it follows that $xy \leq x^p/p + y^q/q$. Letting $x = |f|$, $y = |g|$, and integrating

$$\int |fg|\, d\mu \leq \frac{1}{p} + \frac{1}{q} = 1 = \|f\|_p\|g\|_q$$ □

Remark The special case $p = q = 2$ is called the **Cauchy-Schwarz inequality**. One can give a direct proof of the result in this case by observing that for any θ,

$$0 \leq \int (f + \theta g)^2\, d\mu = \int f^2\, d\mu + \theta\left(2\int fg\, d\mu\right) + \theta^2\left(\int g^2\, d\mu\right)$$

so the quadratic $a\theta^2 + b\theta + c$ on the right-hand side has at most one real root. Recalling the formula for the roots of a quadratic

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

we see $b^2 - 4ac \leq 0$, which is the desired result.

Our next goal is to give conditions that guarantee

$$\lim_{n\to\infty} \int f_n\, d\mu = \int \left(\lim_{n\to\infty} f_n\right) d\mu$$

First, we need a definition. We say that $f_n \to f$ **in measure**, i.e., for any $\epsilon > 0$, $\mu(\{x : |f_n(x) - f(x)| > \epsilon\}) \to 0$ as $n \to \infty$. On a space of finite measure, this is a weaker assumption than $f_n \to f$ a.e., but the next result is easier to prove in the greater generality.

Theorem 1.5.3 (Bounded convergence theorem) *Let E be a set with $\mu(E) < \infty$. Suppose f_n vanishes on E^c, $|f_n(x)| \le M$, and $f_n \to f$ in measure. Then*

$$\int f\,d\mu = \lim_{n\to\infty} \int f_n\,d\mu$$

Example 1.5.4 Consider the real line $\mathbf{R}$ equipped with the Borel sets $\mathcal{R}$ and Lebesgue measure λ. The functions $f_n(x) = 1/n$ on $[0, n]$ and 0 otherwise on show that the conclusion of Theorem 1.5.3 does not hold when $\mu(E) = \infty$.

Proof Let $\epsilon > 0$, $G_n = \{x : |f_n(x) - f(x)| < \epsilon\}$ and $B_n = E - G_n$. Using (iii) and (vi) from Theorem 1.4.7,

$$\begin{aligned}\left|\int f\,d\mu - \int f_n\,d\mu\right| &= \left|\int (f - f_n)\,d\mu\right| \le \int |f - f_n|\,d\mu \\ &= \int_{G_n} |f - f_n|\,d\mu + \int_{B_n} |f - f_n|\,d\mu \\ &\le \epsilon\mu(E) + 2M\mu(B_n)\end{aligned}$$

$f_n \to f$ in measure implies $\mu(B_n) \to 0$. $\epsilon > 0$ is arbitrary and $\mu(E) < \infty$, so the proof is complete. □

Theorem 1.5.5 (Fatou's lemma) *If $f_n \ge 0$, then*

$$\liminf_{n\to\infty} \int f_n\,d\mu \ge \int \left(\liminf_{n\to\infty} f_n\right) d\mu$$

Example 1.5.6 Example 1.5.4 shows that we may have strict inequality in Theorem 1.5.5. The functions $f_n(x) = n1_{(0,1/n]}(x)$ on (0,1) equipped with the Borel sets and Lebesgue measure show that this can happen on a space of finite measure.

Proof Let $g_n(x) = \inf_{m\ge n} f_m(x)$. $f_n(x) \ge g_n(x)$ and as $n \uparrow \infty$,

$$g_n(x) \uparrow g(x) = \liminf_{n\to\infty} f_n(x)$$

Since $\int f_n\,d\mu \ge \int g_n\,d\mu$, it suffices then to show that

$$\liminf_{n\to\infty} \int g_n\,d\mu \ge \int g\,d\mu$$

Let $E_m \uparrow \Omega$ be sets of finite measure. Since $g_n \ge 0$ and for fixed m

$$(g_n \wedge m) \cdot 1_{E_m} \to (g \wedge m) \cdot 1_{E_m} \quad \text{a.e.}$$

the bounded convergence theorem, 1.5.3, implies

$$\liminf_{n\to\infty} \int g_n\,d\mu \ge \int_{E_m} g_n \wedge m\,d\mu \to \int_{E_m} g \wedge m\,d\mu$$

Taking the sup over m and using Theorem 1.4.4 gives the desired result. □

Theorem 1.5.7 (Monotone convergence theorem) *If $f_n \ge 0$ and $f_n \uparrow f$, then*

$$\int f_n\,d\mu \uparrow \int f\,d\mu$$

Proof Fatou's lemma, Theorem 1.5.5, implies $\liminf \int f_n\,d\mu \ge \int f\,d\mu$. On the other hand, $f_n \le f$ implies $\limsup \int f_n\,d\mu \le \int f\,d\mu$. $\square$

Theorem 1.5.8 (Dominated convergence theorem) *If $f_n \to f$ a.e., $|f_n| \le g$ for all n, and g is integrable, then $\int f_n\,d\mu \to \int f\,d\mu$.*

Proof $f_n + g \ge 0$ so Fatou's lemma implies

$$\liminf_{n\to\infty} \int f_n + g\,d\mu \ge \int f + g\,d\mu$$

Subtracting $\int g\,d\mu$ from both sides gives

$$\liminf_{n\to\infty} \int f_n\,d\mu \ge \int f\,d\mu$$

Applying the last result to $-f_n$, we get

$$\limsup_{n\to\infty} \int f_n\,d\mu \le \int f\,d\mu$$

and the proof is complete. $\square$

Exercises

1.5.1 Let $\|f\|_\infty = \inf\{M : \mu(\{x : |f(x)| > M\}) = 0\}$. Prove that

$$\int |fg|d\mu \le \|f\|_1\|g\|_\infty$$

1.5.2 Show that if μ is a probability measure, then

$$\|f\|_\infty = \lim_{p\to\infty} \|f\|_p$$

1.5.3 **Minkowski's inequality.** (i) Suppose $p \in (1,\infty)$. The inequality $|f+g|^p \le 2^p(|f|^p + |g|^p)$ shows that if $\|f\|_p$ and $\|g\|_p$ are $< \infty$, then $\|f+g\|_p < \infty$. Apply Hölder's inequality to $|f||f+g|^{p-1}$ and $|g||f+g|^{p-1}$ to show $\|f+g\|_p \le \|f\|_p + \|g\|_p$. (ii) Show that the last result remains true when $p = 1$ or $p = \infty$.

1.5.4 If f is integrable and E_m are disjoint sets with union E, then

$$\sum_{m=0}^{\infty} \int_{E_m} f\,d\mu = \int_E f\,d\mu$$

So if $f \ge 0$, then $\nu(E) = \int_E f\,d\mu$ defines a measure.

1.5.5 If $g_n \uparrow g$ and $\int g_1^-\,d\mu < \infty$, then $\int g_n\,d\mu \uparrow \int g\,d\mu$.

1.5.6 If $g_m \ge 0$, then $\int \sum_{m=0}^\infty g_m\,d\mu = \sum_{m=0}^\infty \int g_m\,d\mu$.

1.5.7 Let $f \ge 0$. (i) Show that $\int f \wedge n\,d\mu \uparrow \int f\,d\mu$ as $n \to \infty$. (ii) Use (i) to conclude that if g is integrable and $\epsilon > 0$, then we can pick $\delta > 0$ so that $\mu(A) < \delta$ implies $\int_A |g|d\mu < \epsilon$.

1.5.8 Show that if f is integrable on $[a,b]$, $g(x) = \int_{[a,x]} f(y)\,dy$ is continuous on (a,b).

1.5.9 Show that if f has $\|f\|_p = (\int |f|^p d\mu)^{1/p} < \infty$, then there are simple functions φ_n so that $\|\varphi_n - f\|_p \to 0$.

1.5.10 Show that if $\sum_n \int |f_n| d\mu < \infty$, then $\sum_n \int f_n \, d\mu = \int \sum_n f_n \, d\mu$.

1.6 Expected Value

We now specialize to integration with respect to a probability measure P. If $X \geq 0$ is a random variable on $(\Omega, \mathcal{F}, P)$, then we define its **expected value** to be $EX = \int X \, dP$, which always makes sense, but may be ∞. To reduce the general case to the nonnegative case, let $x^+ = \max\{x, 0\}$ be the **positive part** and let $x^- = \max\{-x, 0\}$ be the **negative part** of x. We declare that EX **exists** and set $EX = EX^+ - EX^-$ whenever the subtraction makes sense, i.e., $EX^+ < \infty$ or $EX^- < \infty$.

EX is often called the **mean** of X and denoted by μ. EX is defined by integrating X, so it has all the properties that integrals do. From Theorems 1.4.5 and 1.4.7 and the trivial observation that $E(b) = b$ for any real number b, we get the following:

Theorem 1.6.1 *Suppose $X, Y \geq 0$ or $E|X|, E|Y| < \infty$.*

(a) $E(X + Y) = EX + EY$.

(b) $E(aX + b) = aE(X) + b$ for any real numbers a, b.

(c) If $X \geq Y$, then $EX \geq EY$.

In this section, we will restate some properties of the integral derived in the last section in terms of expected value and prove some new ones. To organize things, we will divide the developments into three subsections.

1.6.1 Inequalities

For probability measures, Theorem 1.5.1 becomes:

Theorem 1.6.2 (Jensen's inequality) *Suppose φ is* **convex***, that is,*

$$\lambda\varphi(x) + (1 - \lambda)\varphi(y) \geq \varphi(\lambda x + (1 - \lambda)y)$$

for all $\lambda \in (0, 1)$ and $x, y \in \mathbf{R}$. Then

$$E(\varphi(X)) \geq \varphi(EX)$$

provided both expectations exist, i.e., $E|X|$ and $E|\varphi(X)| < \infty$.

To recall the direction in which the inequality goes note that if $P(X = x) = \lambda$ and $P(X = y) = 1 - \lambda$, then

$$E\varphi(X) = \lambda\varphi(x) + (1 - \lambda)\varphi(y) \geq \varphi(\lambda x + (1 - \lambda)y) = \varphi(EX)$$

Two useful special cases are $|EX| \leq E|X|$ and $(EX)^2 \leq E(X^2)$.

Theorem 1.6.3 (Hölder's inequality) *If $p, q \in [1, \infty]$ with $1/p + 1/q = 1$, then*

$$E|XY| \leq \|X\|_p \|Y\|_q$$

Here $\|X\|_r = (E|X|^r)^{1/r}$ for $r \in [1, \infty)$; $\|X\|_\infty = \inf\{M : P(|X| > M) = 0\}$.

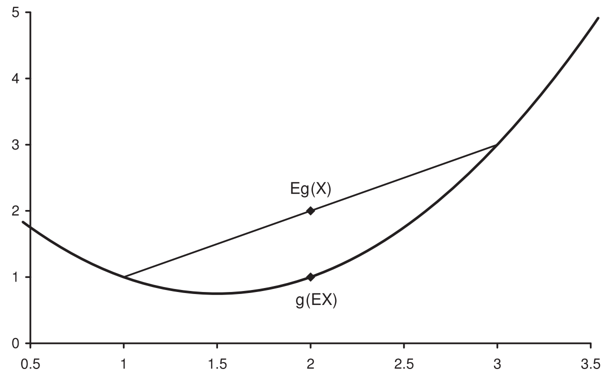

Figure 1.6 Jensen's inequality for $g(x) = x^2 - 3x + 3$, $P(X = 1) = P(X = 3) = 1/2$.

To state our next result, we need some notation. If we only integrate over $A \subset \Omega$, we write

$$E(X; A) = \int_A X\, dP$$

Theorem 1.6.4 (Chebyshev's inequality) *Suppose* $\varphi : \mathbf{R} \to \mathbf{R}$ *has* $\varphi \geq 0$, *let* $A \in \mathcal{R}$ *and let* $i_A = \inf\{\varphi(y) : y \in A\}$.

$$i_A P(X \in A) \leq E(\varphi(X); X \in A) \leq E\varphi(X)$$

Proof The definition of i_A and the fact that $\varphi \geq 0$ imply that

$$i_A 1_{(X \in A)} \leq \varphi(X) 1_{(X \in A)} \leq \varphi(X)$$

So taking expected values and using part (c) of Theorem 1.6.1 gives the desired result. □

Remark Some authors call this result **Markov's inequality** and use the name Chebyshev's inequality for the special case in which $\varphi(x) = x^2$ and $A = \{x : |x| \geq a\}$:

$$a^2 P(|X| \geq a) \leq EX^2 \tag{1.6.1}$$

1.6.2 Integration to the Limit

Our next step is to restate the three classic results from the previous section about what happens when we interchange limits and integrals.

Theorem 1.6.5 (Fatou's lemma) *If* $X_n \geq 0$, *then*

$$\liminf_{n\to\infty} EX_n \geq E(\liminf_{n\to\infty} X_n)$$

Theorem 1.6.6 (Monotone convergence theorem) *If* $0 \leq X_n \uparrow X$, *then* $EX_n \uparrow EX$.

Theorem 1.6.7 (Dominated convergence theorem) *If* $X_n \to X$ *a.s.*, $|X_n| \leq Y$ *for all n, and* $EY < \infty$, *then* $EX_n \to EX$.

The special case of Theorem 1.6.7 in which Y is constant is called the **bounded convergence theorem**.

In the developments that follow, we will need another result on integration to the limit. Perhaps the most important special case of this result occurs when $g(x) = |x|^p$ with $p > 1$ and $h(x) = x$.

Theorem 1.6.8 *Suppose $X_n \to X$ a.s. Let g, h be continuous functions with*

(i) $g \geq 0$ and $g(x) \to \infty$ as $|x| \to \infty$,

(ii) $|h(x)|/g(x) \to 0$ as $|x| \to \infty$,

and (iii) $Eg(X_n) \leq K < \infty$ for all n.

Then $Eh(X_n) \to Eh(X)$.

Proof By subtracting a constant from h, we can suppose without loss of generality that $h(0) = 0$. Pick M large so that $P(|X| = M) = 0$ and $g(x) > 0$ when $|x| \geq M$. Given a random variable Y, let $\bar{Y} = Y1_{(|Y| \leq M)}$. Since $P(|X| = M) = 0$, $\bar{X}_n \to \bar{X}$ a.s. Since $h(\bar{X}_n)$ is bounded and h is continuous, it follows from the bounded convergence theorem that

$$\text{(a)} \qquad Eh(\bar{X}_n) \to Eh(\bar{X})$$

To control the effect of the truncation, we use the following:

$$\text{(b)} \qquad |Eh(\bar{Y}) - Eh(Y)| \leq E|h(\bar{Y}) - h(Y)| \leq E(|h(Y)|; |Y| > M) \leq \epsilon_M Eg(Y)$$

where $\epsilon_M = \sup\{|h(x)|/g(x) : |x| \geq M\}$. To check the second inequality, note that when $|Y| \leq M$, $\bar{Y} = Y$, and we have supposed $h(0) = 0$. The third inequality follows from the definition of ϵ_M.

Taking $Y = X_n$ in (b) and using (iii), it follows that

$$\text{(c)} \qquad |Eh(\bar{X}_n) - Eh(X_n)| \leq K\epsilon_M$$

To estimate $|Eh(\bar{X}) - Eh(X)|$, we observe that $g \geq 0$ and g is continuous, so Fatou's lemma implies

$$Eg(X) \leq \liminf_{n\to\infty} Eg(X_n) \leq K$$

Taking $Y = X$ in (b) gives

$$\text{(d)} \qquad |Eh(\bar{X}) - Eh(X)| \leq K\epsilon_M$$

The triangle inequality implies

$$|Eh(X_n) - Eh(X)| \leq |Eh(X_n) - Eh(\bar{X}_n)| + |Eh(\bar{X}_n) - Eh(\bar{X})| + |Eh(\bar{X}) - Eh(X)|$$

Taking limits and using (a), (c), (d), we have

$$\limsup_{n\to\infty} |Eh(X_n) - Eh(X)| \leq 2K\epsilon_M$$

which proves the desired result since $K < \infty$ and $\epsilon_M \to 0$ as $M \to \infty$. □

1.6.3 Computing Expected Values

Integrating over $(\Omega, \mathcal{F}, P)$ is nice in theory, but to do computations we have to shift to a space on which we can do calculus. In most cases, we will apply the next result with $S = \mathbf{R}^d$.

Theorem 1.6.9 (Change of variables formula) *Let X be a random element of $(S, \mathcal{S})$ with distribution μ, i.e., $\mu(A) = P(X \in A)$. If f is a measurable function from $(S, \mathcal{S})$ to $(\mathbf{R}, \mathcal{R})$ so that $f \geq 0$ or $E|f(X)| < \infty$, then*

$$Ef(X) = \int_S f(y)\, \mu(dy)$$

Remark To explain the name, write h for X and $P \circ h^{-1}$ for μ to get

$$\int_\Omega f(h(\omega))\, dP = \int_S f(y)\, d(P \circ h^{-1})$$

Proof We will prove this result by verifying it in four increasingly more general special cases that parallel the way that the integral was defined in Section 1.4. The reader should note the method employed, since it will be used several times.

CASE 1: INDICATOR FUNCTIONS. If $B \in \mathcal{S}$ and $f = 1_B$, then recalling the relevant definitions shows

$$E 1_B(X) = P(X \in B) = \mu(B) = \int_S 1_B(y)\, \mu(dy)$$

CASE 2: SIMPLE FUNCTIONS. Let $f(x) = \sum_{m=1}^n c_m 1_{B_m}$ where $c_m \in \mathbf{R}$, $B_m \in \mathcal{S}$. The linearity of expected value, the result of Case 1, and the linearity of integration imply

$$\begin{aligned} Ef(X) &= \sum_{m=1}^n c_m E 1_{B_m}(X) \\ &= \sum_{m=1}^n c_m \int_S 1_{B_m}(y)\, \mu(dy) = \int_S f(y)\, \mu(dy) \end{aligned}$$

CASE 3: NONNEGATIVE FUNCTIONS. Now if $f \geq 0$ and we let

$$f_n(x) = ([2^n f(x)]/2^n) \wedge n$$

where $[x] =$ the largest integer $\leq x$ and $a \wedge b = \min\{a, b\}$, then the f_n are simple and $f_n \uparrow f$, so using the result for simple functions and the monotone convergence theorem:

$$Ef(X) = \lim_n Ef_n(X) = \lim_n \int_S f_n(y)\, \mu(dy) = \int_S f(y)\, \mu(dy)$$

CASE 4: INTEGRABLE FUNCTIONS. The general case now follows by writing $f(x) = f(x)^+ - f(x)^-$. The condition $E|f(X)| < \infty$ guarantees that $Ef(X)^+$ and $Ef(X)^-$ are finite. So using the result for nonnegative functions and linearity of expected value and integration:

$$Ef(X) = Ef(X)^+ - Ef(X)^- = \int_S f(y)^+ \, \mu(dy) - \int_S f(y)^- \, \mu(dy)$$
$$= \int_S f(y) \, \mu(dy)$$

which completes the proof. □

A consequence of Theorem 1.6.9 is that we can compute expected values of functions of random variables by performing integrals on the real line. Before we can treat some examples, we need to introduce the terminology for what we are about to compute. If k is a positive integer, then EX^k is called the **kth moment** of X. The first moment EX is usually called the **mean** and denoted by μ. If $EX^2 < \infty$, then the **variance** of X is defined to be $\operatorname{var}(X) = E(X - \mu)^2$. To compute the variance the following formula is useful:

$$\begin{aligned} \operatorname{var}(X) &= E(X - \mu)^2 \\ &= EX^2 - 2\mu EX + \mu^2 = EX^2 - \mu^2 \end{aligned} \tag{1.6.2}$$

From this it is immediate that

$$\operatorname{var}(X) \le EX^2 \tag{1.6.3}$$

Here EX^2 is the expected value of X^2. When we want the square of EX, we will write $(EX)^2$. Since $E(aX+b) = aEX+b$ by (b) of Theorem 1.6.1, it follows easily from the definition that

$$\begin{aligned} \operatorname{var}(aX + b) &= E(aX + b - E(aX + b))^2 \\ &= a^2 E(X - EX)^2 = a^2 \operatorname{var}(X) \end{aligned} \tag{1.6.4}$$

We turn now to concrete examples and leave the calculus in the first two examples to the reader. (Integrate by parts.)

Example 1.6.10 If X has an **exponential distribution** with rate 1, then

$$EX^k = \int_0^\infty x^k e^{-x} dx = k!$$

So the mean of X is 1 and variance is $EX^2 - (EX)^2 = 2 - 1^2 = 1$. If we let $Y = X/\lambda$, then by Exercise 1.2.5, Y has density $\lambda e^{-\lambda y}$ for $y \ge 0$, the **exponential density** with parameter λ. From (b) of Theorem 1.6.1 and (1.6.4), it follows that Y has mean $1/\lambda$ and variance $1/\lambda^2$.

Example 1.6.11 If X has a **standard normal distribution**,

$$EX = \int x(2\pi)^{-1/2} \exp(-x^2/2) \, dx = 0 \quad \text{(by symmetry)}$$
$$\operatorname{var}(X) = EX^2 = \int x^2 (2\pi)^{-1/2} \exp(-x^2/2) \, dx = 1$$

If we let $\sigma > 0$, $\mu \in \mathbf{R}$, and $Y = \sigma X + \mu$, then (b) of Theorem 1.6.1 and (1.6.4), imply $EY = \mu$ and $\operatorname{var}(Y) = \sigma^2$. By Exercise 1.2.5, Y has density

$$(2\pi\sigma^2)^{-1/2} \exp(-(y - \mu)^2/2\sigma^2)$$

the **normal distribution** with mean μ and variance σ^2.

We will next consider some discrete distributions. The first is very simple, but will be useful several times, so we record it here.

Example 1.6.12 We say that X has a **Bernoulli distribution** with parameter p if $P(X = 1) = p$ and $P(X = 0) = 1 - p$. Clearly,

$$EX = p \cdot 1 + (1 - p) \cdot 0 = p$$

Since $X^2 = X$, we have $EX^2 = EX = p$ and

$$\text{var}\,(X) = EX^2 - (EX)^2 = p - p^2 = p(1 - p)$$

Example 1.6.13 We say that X has a **Poisson distribution** with parameter λ if

$$P(X = k) = e^{-\lambda}\lambda^k/k! \text{ for } k = 0, 1, 2, \ldots$$

To evaluate the moments of the Poisson random variable, we use a little inspiration to observe that for $k \geq 1$

$$\begin{aligned} E(X(X-1)\cdots(X-k+1)) &= \sum_{j=k}^{\infty} j(j-1)\cdots(j-k+1)e^{-\lambda}\frac{\lambda^j}{j!} \\ &= \lambda^k \sum_{j=k}^{\infty} e^{-\lambda}\frac{\lambda^{j-k}}{(j-k)!} = \lambda^k \end{aligned}$$

where the equalities follow from (i) the facts that $j(j-1)\cdots(j-k+1) = 0$ when $j < k$, (ii) cancelling part of the factorial, and (iii) the fact that Poisson distribution has total mass 1. Using the last formula, it follows that $EX = \lambda$ while

$$\text{var}\,(X) = EX^2 - (EX)^2 = E(X(X-1)) + EX - \lambda^2 = \lambda$$

Example 1.6.14 N is said to have a **geometric distribution** with success probability $p \in (0, 1)$ if

$$P(N = k) = p(1 - p)^{k-1} \quad \text{for } k = 1, 2, \ldots$$

N is the number of independent trials needed to observe an event with probability p. Differentiating the identity

$$\sum_{k=0}^{\infty}(1 - p)^k = 1/p$$

and referring to Example A.5.6 for the justification gives

$$\begin{aligned} -\sum_{k=1}^{\infty} k(1-p)^{k-1} &= -1/p^2 \\ \sum_{k=2}^{\infty} k(k-1)(1-p)^{k-2} &= 2/p^3 \end{aligned}$$

From this it follows that

$$EN = \sum_{k=1}^{\infty} kp(1-p)^{k-1} = 1/p$$

$$EN(N-1) = \sum_{k=1}^{\infty} k(k-1)p(1-p)^{k-1} = 2(1-p)/p^2$$

$$\begin{aligned} \text{var}\,(N) &= EN^2 - (EN)^2 = EN(N-1) + EN - (EN)^2 \\ &= \frac{2(1-p)}{p^2} + \frac{p}{p^2} - \frac{1}{p^2} = \frac{1-p}{p^2} \end{aligned}$$

Exercises

1.6.1 Suppose φ is strictly convex, i.e., $>$ holds for $\lambda \in (0,1)$. Show that, under the assumptions of Theorem 1.6.2, $\varphi(EX) = E\varphi(X)$ implies $X = EX$ a.s.

1.6.2 Suppose $\varphi : \mathbf{R}^n \to \mathbf{R}$ is convex. Imitate the proof of Theorem 1.5.1 to show

$$E\varphi(X_1, \dots, X_n) \geq \varphi(EX_1, \dots, EX_n)$$

provided $E|\varphi(X_1, \dots, X_n)| < \infty$ and $E|X_i| < \infty$ for all i.

1.6.3 **Chebyshev's inequality is and is not sharp.** (i) Show that Theorem 1.6.4 is sharp by showing that if $0 < b \leq a$ are fixed there is an X with $EX^2 = b^2$ for which $P(|X| \geq a) = b^2/a^2$. (ii) Show that Theorem 1.6.4 is not sharp by showing that if X has $0 < EX^2 < \infty$, then

$$\lim_{a \to \infty} a^2 P(|X| \geq a)/EX^2 = 0$$

1.6.4 **One-sided Chebyshev bound.** (i) Let $a > b > 0$, $0 < p < 1$, and let X have $P(X = a) = p$ and $P(X = -b) = 1 - p$. Apply Theorem 1.6.4 to $\varphi(x) = (x + b)^2$ and conclude that if Y is any random variable with $EY = EX$ and $\text{var}\,(Y) = \text{var}\,(X)$, then $P(Y \geq a) \leq p$ and equality holds when $Y = X$.
(ii) Suppose $EY = 0$, $\text{var}\,(Y) = \sigma^2$, and $a > 0$. Show that $P(Y \geq a) \leq \sigma^2/(a^2 + \sigma^2)$, and there is a Y for which equality holds.

1.6.5 **Two nonexistent lower bounds.**
Show that: (i) if $\epsilon > 0$, $\inf\{P(|X| > \epsilon) : EX = 0, \text{var}\,(X) = 1\} = 0$.
(ii) if $y \geq 1$, $\sigma^2 \in (0, \infty)$, $\inf\{P(|X| > y) : EX = 1, \text{var}\,(X) = \sigma^2\} = 0$.

1.6.6 **A useful lower bound.** Let $Y \geq 0$ with $EY^2 < \infty$. Apply the Cauchy-Schwarz inequality to $Y 1_{(Y>0)}$ and conclude

$$P(Y > 0) \geq (EY)^2/EY^2$$

1.6.7 Let $\Omega = (0,1)$ equipped with the Borel sets and Lebesgue measure. Let $\alpha \in (1,2)$ and $X_n = n^\alpha 1_{(1/(n+1), 1/n)} \to 0$ a.s. Show that Theorem 1.6.8 can be applied with $h(x) = x$ and $g(x) = |x|^{2/\alpha}$, but the X_n are not dominated by an integrable function.

1.6.8 Suppose that the probability measure μ has $\mu(A) = \int_A f(x)\,dx$ for all $A \in \mathcal{R}$. Use the proof technique of Theorem 1.6.9 to show that for any g with $g \geq 0$ or $\int |g(x)|\,\mu(dx) < \infty$ we have

$$\int g(x)\,\mu(dx) = \int g(x) f(x)\,dx$$

1.6.9 **Inclusion-exclusion formula**. Let $A_1, A_2, \ldots A_n$ be events and $A = \cup_{i=1}^n A_i$. Prove that $1_A = 1 - \prod_{i=1}^n (1 - 1_{A_i})$. Expand out the right-hand side, then take expected value to conclude

$$\begin{aligned} P\left(\cup_{i=1}^n A_i\right) &= \sum_{i=1}^n P(A_i) - \sum_{i<j} P(A_i \cap A_j) \\ &\quad + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n-1} P(\cap_{i=1}^n A_i) \end{aligned}$$

1.6.10 **Bonferroni inequalities.** Let $A_1, A_2, \ldots A_n$ be events and $A = \cup_{i=1}^n A_i$. Show that $1_A \leq \sum_{i=1}^n 1_{A_i}$, etc. and then take expected values to conclude

$$P\left(\cup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i)$$

$$P\left(\cup_{i=1}^n A_i\right) \geq \sum_{i=1}^n P(A_i) - \sum_{i<j} P(A_i \cap A_j)$$

$$P\left(\cup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k)$$

In general, if we stop the inclusion–exclusion formula after an even (odd) number of sums, we get a(n) lower (upper) bound.

1.6.11 If $E|X|^k < \infty$, then for $0 < j < k$, $E|X|^j < \infty$, and furthermore

$$E|X|^j \leq (E|X|^k)^{j/k}$$

1.6.12 Apply Jensen's inequality with $\varphi(x) = e^x$ and $P(X = \log\, y_m) = p(m)$ to conclude that if $\sum_{m=1}^n p(m) = 1$ and $p(m), y_m > 0$, then

$$\sum_{m=1}^n p(m) y_m \geq \prod_{m=1}^n y_m^{p(m)}$$

When $p(m) = 1/n$, this says the arithmetic mean exceeds the geometric mean.

1.6.13 If $EX_1^- < \infty$ and $X_n \uparrow X$, then $EX_n \uparrow EX$.

1.6.14 Let $X \geq 0$ but do NOT assume $E(1/X) < \infty$. Show

$$\lim_{y\to\infty} yE(1/X; X > y) = 0, \qquad \lim_{y\downarrow 0} yE(1/X; X > y) = 0.$$

1.6.15 If $X_n \geq 0$, then $E(\sum_{n=0}^\infty X_n) = \sum_{n=0}^\infty EX_n$.

1.6.16 If X is integrable and A_n are disjoint sets with union A, then

$$\sum_{n=0}^{\infty} E(X; A_n) = E(X; A)$$

i.e., the sum converges absolutely and has the value on the right.

1.7 Product Measures, Fubini's Theorem

Let $(X, \mathcal{A}, \mu_1)$ and $(Y, \mathcal{B}, \mu_2)$ be two σ-finite measure spaces. Let

$$\Omega = X \times Y = \{(x, y) : x \in X, y \in Y\}$$
$$\mathcal{S} = \{A \times B : A \in \mathcal{A}, B \in \mathcal{B}\}$$

Sets in $\mathcal{S}$ are called **rectangles**. It is easy to see that $\mathcal{S}$ is a semi-algebra:

$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$$
$$(A \times B)^c = (A^c \times B) \cup (A \times B^c) \cup (A^c \times B^c)$$

Let $\mathcal{F} = \mathcal{A} \times \mathcal{B}$ be the σ-algebra generated by $\mathcal{S}$.

Theorem 1.7.1 *There is a unique measure μ on $\mathcal{F}$ with*

$$\mu(A \times B) = \mu_1(A)\mu_2(B)$$

Notation μ is often denoted by $\mu_1 \times \mu_2$.

Proof By Theorem 1.1.9 it is enough to show that if $A \times B = +_i (A_i \times B_i)$ is a finite or countable disjoint union, then

$$\mu(A \times B) = \sum_i \mu(A_i \times B_i)$$

For each $x \in A$, let $I(x) = \{i : x \in A_i\}$. $B = +_{i \in I(x)} B_i$, so

$$1_A(x)\mu_2(B) = \sum_i 1_{A_i}(x)\mu_2(B_i)$$

Integrating with respect to μ_1 and using Exercise 1.5.6 gives

$$\mu_1(A)\mu_2(B) = \sum_i \mu_1(A_i)\mu_2(B_i)$$

which proves the result. □

Using Theorem 1.7.1 and induction, it follows that if $(\Omega_i, \mathcal{F}_i, \mu_i)$, $i = 1, \ldots, n$, are σ-finite measure spaces and $\Omega = \Omega_1 \times \cdots \times \Omega_n$, there is a unique measure μ on the σ-algebra $\mathcal{F}$ generated by sets of the form $A_1 \times \cdots \times A_n$, $A_i \in \mathcal{F}_i$, which has

$$\mu(A_1 \times \cdots \times A_n) = \prod_{m=1}^{n} \mu_m(A_m)$$

When $(\Omega_i, \mathcal{F}_i, \mu_i) = (\mathbf{R}, \mathcal{R}, \lambda)$ for all i, the result is Lebesgue measure on the Borel subsets of n dimensional Euclidean space $\mathbf{R}^n$.

Returning to the case in which $(\Omega, \mathcal{F}, \mu)$ is the product of two measure spaces, $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$, our next goal is to prove:

Theorem 1.7.2 (Fubini's theorem) *If $f \geq 0$ or $\int |f|\, d\mu < \infty$, then*

$$(*) \qquad \int_X \int_Y f(x,y)\, \mu_2(dy)\, \mu_1(dx) = \int_{X \times Y} f\, d\mu = \int_Y \int_X f(x,y)\, \mu_1(dx)\, \mu_2(dy)$$

Proof We will prove only the first equality, since the second follows by symmetry. Two technical things that need to be proved before we can assert that the first integral makes sense are:

When x is fixed, $y \to f(x,y)$ is $\mathcal{B}$ measurable.

$x \to \int_Y f(x,y)\mu_2(dy)$ is $\mathcal{A}$ measurable.

We begin with the case $f = 1_E$. Let $E_x = \{y : (x,y) \in E\}$ be the **cross-section** at x.

Lemma 1.7.3 *If $E \in \mathcal{F}$, then $E_x \in \mathcal{B}$.*

Proof $(E^c)_x = (E_x)^c$ and $(\cup_i E_i)_x = \cup_i (E_i)_x$, so if $\mathcal{E}$ is the collection of sets E for which $E_x \in \mathcal{B}$, then $\mathcal{E}$ is a σ-algebra. Since $\mathcal{E}$ contains the rectangles, the result follows. □

Lemma 1.7.4 *If $E \in \mathcal{F}$, then $g(x) \equiv \mu_2(E_x)$ is $\mathcal{A}$ measurable and*

$$\int_X g\, d\mu_1 = \mu(E)$$

Notice that it is not obvious that the collection of sets for which the conclusion is true is a σ-algebra since $\mu(E_1 \cup E_2) = \mu(E_1) + \mu(E_2) - \mu(E_1 \cap E_2)$. Dynkin's $\pi - \lambda$ Theorem (A.1.4) was tailor-made for situations like this.

Proof If conclusions hold for E_n and $E_n \uparrow E$, then Theorem 1.3.7 and the monotone convergence theorem imply that they hold for E. Since μ_1 and μ_2 are σ-finite, it is enough then to prove the result for $E \subset F \times G$ with $\mu_1(F) < \infty$ and $\mu_2(G) < \infty$, or taking $\Omega = F \times G$ we can suppose without loss of generality that $\mu(\Omega) < \infty$. Let $\mathcal{L}$ be the collection of sets E for which the conclusions hold. We will now check that $\mathcal{L}$ is a λ-system. Property (i) of a λ-system is trivial. (iii) follows from the first sentence in the proof. To check (ii) we observe that

$$\mu_2((A - B)_x) = \mu_2(A_x - B_x) = \mu_2(A_x) - \mu_2(B_x)$$

and integrating over x gives the second conclusion. Since $\mathcal{L}$ contains the rectangles, a π-system that generates $\mathcal{F}$, the desired result follows from the $\pi - \lambda$ theorem. □

We are now ready to prove Theorem 1.7.2 by verifying it in four increasingly more general special cases.

Case 1. If $E \in \mathcal{F}$ and $f = 1_E$, then $(*)$ follows from Lemma 1.7.4.

Case 2. Since each integral is linear in f, it follows that $(*)$ holds for simple functions.

Case 3. Now if $f \geq 0$ and we let $f_n(x) = ([2^n f(x)]/2^n) \wedge n$, where $[x] =$ the largest integer $\leq x$, then the f_n are simple and $f_n \uparrow f$, so it follows from the monotone convergence theorem that $(*)$ holds for all $f \geq 0$.

Case 4. The general case now follows by writing $f(x) = f(x)^+ - f(x)^-$ and applying Case 3 to f^+, f^-, and $|f|$. □

To illustrate why the various hypotheses of Theorem 1.7.2 are needed, we will now give some examples where the conclusion fails.

Example 1.7.5 Let $X = Y = \{1, 2, \ldots\}$ with $\mathcal{A} = \mathcal{B} =$ all subsets and $\mu_1 = \mu_2 =$ counting measure. For $m \geq 1$, let $f(m,m) = 1$ and $f(m+1,m) = -1$, and let $f(m,n) = 0$ otherwise. We claim that

$$\sum_m \sum_n f(m,n) = 1 \quad \text{but} \quad \sum_n \sum_m f(m,n) = 0$$

A picture is worth several dozen words:

$$\begin{array}{cccccc}
 & \vdots & \vdots & \vdots & \vdots & \\
 & 0 & 0 & 0 & 1 & \ldots \\
\uparrow & 0 & 0 & 1 & -1 & \ldots \\
n & 0 & 1 & -1 & 0 & \ldots \\
 & 1 & -1 & 0 & 0 & \ldots \\
 & & m & \to & &
\end{array}$$

In words, if we sum the columns first, the first one gives us a 1 and the others 0, while if we sum the rows each one gives us a 0.

Example 1.7.6 Let $X = (0,1)$, $Y = (1,\infty)$, both equipped with the Borel sets and Lebesgue measure. Let $f(x,y) = e^{-xy} - 2e^{-2xy}$.

$$\int_0^1 \int_1^\infty f(x,y)\,dy\,dx = \int_0^1 x^{-1}(e^{-x} - e^{-2x})\,dx > 0$$
$$\int_1^\infty \int_0^1 f(x,y)\,dx\,dy = \int_1^\infty y^{-1}(e^{-2y} - e^{-y})\,dy < 0$$

The next example indicates why μ_1 and μ_2 must be σ-finite.

Example 1.7.7 Let $X = (0,1)$ with $\mathcal{A} =$ the Borel sets and $\mu_1 =$ Lebesgue measure. Let $Y = (0,1)$ with $\mathcal{B} =$ all subsets and $\mu_2 =$ counting measure. Let $f(x,y) = 1$ if $x = y$ and 0 otherwise

$$\int_Y f(x,y)\,\mu_2(dy) = 1 \quad \text{for all } x \text{ so} \quad \int_X \int_Y f(x,y)\,\mu_2(dy)\,\mu_1(dx) = 1$$
$$\int_X f(x,y)\,\mu_1(dx) = 0 \quad \text{for all } y \text{ so} \quad \int_Y \int_X f(x,y)\,\mu_1(dy)\,\mu_2(dx) = 0$$

Our last example shows that measurability is important or maybe that some of the axioms of set theory are not as innocent as they seem.

Example 1.7.8 By the axiom of choice and the continuum hypothesis one can define an order relation $<'$ on (0,1) so that $\{x : x <' y\}$ is countable for each y. Let $X = Y = (0,1)$, let

$\mathcal{A} = \mathcal{B} =$ the Borel sets and $\mu_1 = \mu_2 =$ Lebesgue measure. Let $f(x,y) = 1$ if $x <' y$, $= 0$ otherwise. Since $\{x : x <' y\}$ and $\{y : x <' y\}^c$ are countable,

$$\int_X f(x,y)\,\mu_1(dx) = 0 \quad \text{for all } y$$
$$\int_Y f(x,y)\,\mu_2(dy) = 1 \quad \text{for all } x$$

Exercises

1.7.1 If $\int_X \int_Y |f(x,y)|\mu_2(dy)\mu_1(dx) < \infty$, then

$$\int_X \int_Y f(x,y)\mu_2(dy)\mu_1(dx) = \int_{X\times Y} f\,d(\mu_1 \times \mu_2) = \int_Y \int_X f(x,y)\mu_1(dx)\mu_2(dy)$$

Corollary Let $X = \{1, 2, \ldots\}$, $\mathcal{A} =$ all subsets of X, and $\mu_1 =$ counting measure. If $\sum_n \int |f_n| d\mu < \infty$, then $\sum_n \int f_n\,d\mu = \int \sum_n f_n\,d\mu$.

1.7.2 Let $g \geq 0$ be a measurable function on $(X, \mathcal{A}, \mu)$. Use Theorem 1.7.2 to conclude that

$$\int_X g\,d\mu = (\mu \times \lambda)(\{(x,y) : 0 \leq y < g(x)\}) = \int_0^\infty \mu(\{x : g(x) > y\})\,dy$$

1.7.3 Let F, G be Stieltjes measure functions and let μ, ν be the corresponding measures on $(\mathbf{R}, \mathcal{R})$. Show that

(i) $\int_{(a,b]} \{F(y) - F(a)\} dG(y) = (\mu \times \nu)(\{(x,y) : a < x \leq y \leq b\})$

(ii) $\int_{(a,b]} F(y)\,dG(y) + \int_{(a,b]} G(y)\,dF(y)$

$$= F(b)G(b) - F(a)G(a) + \sum_{x \in (a,b]} \mu(\{x\})\nu(\{x\})$$

(iii) If $F = G$ is continuous, then $\int_{(a,b]} 2F(y) dF(y) = F^2(b) - F^2(a)$.

To see the second term in (ii) is needed, let $F(x) = G(x) = 1_{[0,\infty)}(x)$ and $a < 0 < b$.

1.7.4 Let μ be a finite measure on $\mathbf{R}$ and $F(x) = \mu((-\infty, x])$. Show that

$$\int (F(x+c) - F(x))\,dx = c\mu(\mathbf{R})$$

1.7.5 Show that $e^{-xy}\sin x$ is integrable in the strip $0 < x < a$, $0 < y$. Perform the double integral in the two orders to get:

$$\int_0^a \frac{\sin x}{x}\,dx = \arctan(a) - (\cos a)\int_0^\infty \frac{e^{-ay}}{1+y^2}\,dy - (\sin a)\int_0^\infty \frac{ye^{-ay}}{1+y^2}\,dy$$

and replace $1 + y^2$ by 1 to conclude $\left|\int_0^a (\sin x)/x\,dx - \arctan(a)\right| \leq 2/a$ for $a \geq 1$.